Kick-drift LSVD — manifolds, crossing ages and the distribution of S

annealed 10 seeds, quenched 10 seeds, cosine 10 seeds; 60 cycles; Python engine (python/kd_engine.py), multi-seed, mean ± sd over seeds. Seed 0 reproduces the MATLAB rng(0) run exactly.

Summary

• All PASS/FAIL checks pass (table below): the age bookkeeping reproduces the existing re-entry and exit counts exactly, the pooled age-1 re-entry hazard equals Type 1, ρ from 1/|a| matches the histogram within 2% away from caustics, and adding Parts 1/2/4 to the MATLAB script leaves every existing output bit-identical (Octave, annealed, 13 cycles). The MATLAB port gives the same numbers as Python on identical kicks.

• Kaplan log-normal tail test (4C), reproduced first: annealed predicted $\alpha^2/2v = 0.0724$ vs measured 0.1088 (seed 0 earlier: 0.072 vs 0.10–0.11), quenched 0.0319 vs 0.0266 (earlier 0.032 vs 0.026). Over 10 seeds: annealed 0.0717 ± 0.0014 vs 0.1085 ± 0.0018 — the low-S tail decays 1.5× FASTER than the log-normal predicts; quenched 0.0370 ± 0.0061 vs 0.0332 ± 0.0070 (0.90×); cosine 0.0125 vs 0.0210.

• Why annealed misses the log-normal (4D): the annealed run is, parcel by parcel, an i.i.d. product of step Jacobians. The i.i.d. model (10^7 samples) reproduces the growth rates and the decay of P(S ≤ 2) (0.1090 vs 0.1085 ± 0.0018), with P(S ≤ s, N) within 2.5 standard errors of the 10-seed mean (ratio 0.96 at N = 20, 0.98 at N = 60; the i.i.d. values run 2–7% low at N ≥ 40). The generalised Lyapunov exponent has the plateau large-deviation theory predicts for a quantity bounded below (σmax ≥ 1): L(q) is flat to 10^{-3} for q ≤ −2 at −0.101, so −min L(q) ≈ I(0) ≈ 0.10–0.11, while the Gaussian $\alpha^2/2v = 0.072$ sits at the centre of the distribution and underestimates the far lower tail. Near S = 1, P(S ≤ s) ∝ $(\ln s)^2$ (σmax = 1 is a codimension-2 set): measured P(1.25)/P(2) = 0.097 against $(\ln 1.25/\ln 2)^2 = 0.104$.

• Caustics (1A): the exact count on the row0 line grows at 0.242 ± 0.009 /cycle (annealed), its exact line length at 0.242, against Kaplan Eq. (6) α + β/4 = 0.234 ± 0.006 — Eq. (6) holds. Quenched: 0.198 vs 0.185, consistent within the seed spread. Cosine fails it: caustics 0.207, line length 0.176, but α + β/4 = 0.061. In a mixed phase space most of the line sits in regular islands (r ≈ 0) while the chaotic layer carries the growth, so r is not Gaussian and its mean and variance do not predict E[e^r]. A fixed grid of parcels only gives a lower bound on the count (at 512/px it saturates after N ≈ 12), so the counts come from tracking the line exactly as piecewise-affine pieces.

• Branches and low stretch (1A-iv): across x, the ray density ρ correlates with the local share of S ≤ 2 (annealed r = 0.25 at N = 20) and more strongly with the local mean of −r (r = 0.73): bright branches are where the line is compressed, and S ≥ e^r makes low-S parcels a subset of compressed ones (P(r ≤ ln 2 | S ≤ 2) = 1 exactly; P(S ≤ 2 | r ≤ ln 2) = 0.110 at N = 30).

• Crossing ages (2B): a parcel that leaves the low set comes back within one cycle with probability 0.121 (annealed), 0.128 (quenched), 0.085 (cosine); Kaplan–Meier whole-run return probability 0.283 / 0.330 / 0.366, against naive fractions 0.283 / 0.328 / 0.359 that ignore censoring.

• "Returned before → more likely to return" (2C): crude ratios 1.16 (annealed), 1.77 (quenched; earlier 1.8×), 2.57 (cosine; earlier 2.6×). At fixed depth S_{N-1} the effect largely goes away: in (2, 2.5] the ratio is 0.93 / 1.01 / 0.91, and the Mantel–Haenszel ratio adjusted for depth (7 bins) × age is 0.94 / 1.03 / 1.09, while adjusting for age alone leaves 0.98 / 1.34 / 1.60. The excess is a depth effect — returners sit closer to the threshold — not a memory of having returned.

• Metric (4E): measuring S with y stretched by f = 0.5, 2, 4 leaves α unchanged in every mode (to < 1%). For annealed it moves P(S ≤ 2) by up to 20% but the tail rates by ≤ 3% (0.107 / 0.108 / 0.105 for f = 0.5, 2, 4 vs 0.109), so 4C is metric-robust there. For cosine the rate itself moves (0.022 / 0.014 / 0.014 vs 0.021): its low-S set lies in regular islands where S grows polynomially, and what counts as S ≤ 2 depends on the metric at every N. Thresholds S_THR = 1.5 / 2 / 3 give annealed rates 0.109 / 0.109 / 0.107.

Checks

	check	detail
PASS	Σ age re-entry events = Σ re-entries (haz_reent·count) [annealed, 10 seeds]	421988 = 421988 (seed 0)
PASS	Σ age exit events = Σ exits (haz_gross·count) [annealed]	1343363 = 1343363 (seed 0)
PASS	pooled age-1 re-entry hazard = Type 1 (consecutive masks) [annealed]	0.143448 = 0.143448 (seed 0)
PASS	rho: 1/|a| vs histogram within 2% away from caustics [annealed, seed 0]	4494 bins, median |rel| 0.0041, 96.0% within 2%
PASS	identity S ≥ e^r on every line parcel [annealed]	0 violations
PASS	Σ age re-entry events = Σ re-entries (haz_reent·count) [quenched, 10 seeds]	492522 = 492522 (seed 0)
PASS	Σ age exit events = Σ exits (haz_gross·count) [quenched]	1402530 = 1402530 (seed 0)

	check	detail
PASS	pooled age-1 re-entry hazard = Type 1 (consecutive masks) [quenched]	0.124434 = 0.124434 (seed 0)
PASS	rho: 1/|a| vs histogram within 2% away from caustics [quenched, seed 0]	3139 bins, median |rel| 0.0025, 97.1% within 2%
PASS	identity S ≥ e^r on every line parcel [quenched]	0 violations
PASS	Σ age re-entry events = Σ re-entries (haz_reent·count) [cosine, 10 seeds]	499756 = 499756 (seed 0)
PASS	Σ age exit events = Σ exits (haz_gross·count) [cosine]	1397640 = 1397640 (seed 0)
PASS	pooled age-1 re-entry hazard = Type 1 (consecutive masks) [cosine]	0.084703 = 0.084703 (seed 0)
PASS	rho: 1/|a| vs histogram within 2% away from caustics [cosine, seed 0]	5546 bins, median |rel| 0.0007, 97.8% within 2%
PASS	identity S ≥ e^r on every line parcel [cosine]	0 violations
PASS	MATLAB annealed (Octave, cycles 13): existing cycleData fields and workspace variables unchanged by the port	128/129 hashes identical (54 cycleData fields, 33 metadata, 42 workspace variables); only metadata.codeName differs = the harness script name
PASS	MATLAB port = Python on identical kicks: Part 2 risk/event arrays, Type 1, band projections, exact and grid caustic counts	integer arrays identical
PASS	MATLAB port = Python on identical kicks: rho_hist, rho from 1/|a|, r moments, correlations, 4F	max |diff| 1.2e-14
PASS	MATLAB Part 4 on the 1-px Eulerian lattice = Python window-selected full-period lattice	P(S<=s) and ln S moments agree to 1e-4 at N = 1..13 (independent samples of one field)
PASS	seed 0 = MATLAB rng(0,'twister') stream	RandomState(5489): 0.8147, 0.9058, 0.1270 = MATLAB rng(0); rand(1,3); stream order kick (959) then dot jitter (51200) per cycle
PASS	kick generator transcription (vs Octave on Octave's own stream)	2e-15 given the same spectral filter; the filter itself differs by 1.5e-7 between FFT libraries (sqrt of round-off in the Gaussian tail), so kicks agree to 1e-7 relative
PASS	MATLAB checks printed by the script (Octave run)	re-entry 307803 = 307803; exits 1183657 = 1183657; age-1 = Type 1 0.140992; rho 97.3% within 2%; S >= e^r 0 violations

Part 1 — launched lines of rays

1A. Caustic growth vs Kaplan Eq. (6) prediction α + β/4

Caustics = sign changes of a = ∂x_N/∂x_0 along the line, counted exactly (piecewise-affine line, budget 1.2·10⁷ pieces). α = slope of mean r, β = 2·slope of var r over N = 10..60; growth fitted on ln(count) over N = 10..last exact N.

mode	launch row	caustic growth (fit)	line-length growth (exact)	alpha+beta/4	ratio caustic/pred	alpha	beta	fit N
annealed	row0-rows/8	0.2446 ± 0.0076	0.2433 ± 0.0029	0.2348 ± 0.0036	1.04	0.1537 ± 0.0034	0.3241 ± 0.0198	10–31
annealed	row0	0.2417 ± 0.0085	0.2423 ± 0.0039	0.2344 ± 0.0057	1.03	0.1522 ± 0.0034	0.3290 ± 0.0191	10–31
annealed	row0+rows/8	0.2488 ± 0.0082	0.2468 ± 0.0040	0.2386 ± 0.0058	1.04	0.1520 ± 0.0034	0.3463 ± 0.0270	10–31
quenched	row0-rows/8	0.2162 ± 0.0217	0.2150 ± 0.0144	0.2347 ± 0.0159	0.92	0.1422 ± 0.0153	0.3701 ± 0.0753	10–33
quenched	row0	0.1983 ± 0.0166	0.1919 ± 0.0174	0.1848 ± 0.0376	1.11	0.0668 ± 0.0168	0.4717 ± 0.0962	10–35
quenched	row0+rows/8	0.2246 ± 0.0245	0.2148 ± 0.0165	0.2319 ± 0.0174	0.97	0.1414 ± 0.0197	0.3619 ± 0.0604	10–33
cosine	row0-rows/8	0.2173 ± 0.0000	0.1546 ± 0.0000	0.0404 ± 0.0000	5.38	0.0266 ± 0.0000	0.0552 ± 0.0000	10–44
cosine	row0	0.2066 ± 0.0000	0.1756 ± 0.0000	0.0606 ± 0.0000	3.41	0.0290 ± 0.0000	0.1267 ± 0.0000	10–41
cosine	row0+rows/8	0.2173 ± 0.0000	0.1546 ± 0.0000	0.0404 ± 0.0000	5.38	0.0266 ± 0.0000	0.0552 ± 0.0000	10–44

1A(iv). Correlation across x of the ray density ρ

Bins of 4 px with ≥ 5 parcels; mean over the three launched rows. Spearman uses average ranks.

mode	N	r(rho, frac S≤2) Pearson	Spearman	r(rho, mean −r) Pearson	Spearman
annealed	5	+0.412 ± +0.082	+0.516 ± +0.047	+0.685 ± +0.041	+0.820 ± +0.055
annealed	10	+0.341 ± +0.075	+0.374 ± +0.047	+0.707 ± +0.040	+0.793 ± +0.028
annealed	20	+0.246 ± +0.053	+0.249 ± +0.030	+0.732 ± +0.023	+0.730 ± +0.037
annealed	40	+0.162 ± +0.052	+0.132 ± +0.021	+0.623 ± +0.041	+0.522 ± +0.033
annealed	60	+0.077 ± +0.036	+0.060 ± +0.018	+0.403 ± +0.073	+0.311 ± +0.039
quenched	5	+0.354 ± +0.074	+0.394 ± +0.073	+0.614 ± +0.075	+0.778 ± +0.070
quenched	10	+0.322 ± +0.076	+0.417 ± +0.073	+0.646 ± +0.043	+0.715 ± +0.093
quenched	20	+0.351 ± +0.040	+0.346 ± +0.045	+0.731 ± +0.035	+0.772 ± +0.052
quenched	40	+0.316 ± +0.061	+0.280 ± +0.047	+0.694 ± +0.070	+0.680 ± +0.084
quenched	60	+0.326 ± +0.067	+0.246 ± +0.043	+0.649 ± +0.069	+0.600 ± +0.084
cosine	5	-0.092 ± +0.000	+0.124 ± +0.000	+0.199 ± +0.000	+0.471 ± +0.000
cosine	10	+0.272 ± +0.000	+0.445 ± +0.000	+0.596 ± +0.000	+0.758 ± +0.000
cosine	20	+0.730 ± +0.000	+0.363 ± +0.000	+0.679 ± +0.000	+0.810 ± +0.000
cosine	40	+0.571 ± +0.000	+0.460 ± +0.000	+0.704 ± +0.000	+0.803 ± +0.000
cosine	60	+0.262 ± +0.000	+0.212 ± +0.000	+0.614 ± +0.000	+0.742 ± +0.000

Part 2 — crossing ages

2A. Pooled age-bin hazards

Tracers = the MATLAB lattice (seeded every 2 px in the window). Re-entry: at risk = out of the set at N−1, age = N − exit cycle. Exit: at risk = in the set at N−1, age = N − entry cycle, entrants only (the original cohort is separate in 2B).

mode	age	h_re	at risk (mean)	h_exit (entrants)	at risk (mean)
annealed	1	0.1215 ± 0.0100	1284674	0.4005 ± 0.0098	363268
annealed	2	0.0420 ± 0.0028	1128321	0.3531 ± 0.0071	217848
annealed	3–5	0.0276 ± 0.0019	3138850	0.3856 ± 0.0091	282999
annealed	6–10	0.0093 ± 0.0006	4854539	0.3822 ± 0.0118	76122
annealed	11–20	0.0023 ± 0.0001	9348434	0.4035 ± 0.0320	7119
annealed	21–40	0.0003 ± 0.0000	18207108	0.4591 ± 0.1706	44
annealed	>40	0.0000 ± 0.0000	13235531	nan ± nan	0
quenched	1	0.1276 ± 0.0158	1361529	0.3877 ± 0.0226	447016
quenched	2	0.0572 ± 0.0047	1186068	0.3448 ± 0.0388	272576
quenched	3–5	0.0336 ± 0.0061	3212260	0.3596 ± 0.0361	366806
quenched	6–10	0.0117 ± 0.0017	4887164	0.3369 ± 0.0571	122987
quenched	11–20	0.0030 ± 0.0004	9265501	0.2861 ± 0.0627	21964
quenched	21–40	0.0006 ± 0.0001	17804763	0.0887 ± 0.0470	4370
quenched	>40	0.0002 ± 0.0000	12969410	0.0421 ± 0.0266	691
cosine	1	0.0847 ± 0.0000	1393620	0.4506 ± 0.0000	494520
cosine	2	0.1036 ± 0.0000	1265507	0.6742 ± 0.0000	269372
cosine	3–5	0.0704 ± 0.0000	3018936	0.8201 ± 0.0000	104707
cosine	6–10	0.0035 ± 0.0000	4506631	0.7892 ± 0.0000	854
cosine	11–20	0.0001 ± 0.0000	8820616	nan ± nan	0
cosine	21–40	0.0005 ± 0.0000	17124086	nan ± nan	0
cosine	>40	0.0011 ± 0.0000	12481390	nan ± nan	0

2B. Kaplan–Meier F(k) and the naive fraction

F(k) = 1 − Π_{j≤k}(1 − h(j)), hazards with < 200 at risk dropped; "whole run" = largest k with ≥ 200 at risk. Naive = share of spells that ended by the end of the run (censoring ignored).

mode	direction	F(1)	F(2)	F(3)	F(5)	F(10)	F(20)	F(40)	whole run	naive
annealed	re-entry	0.121 ± 0.010	0.158 ± 0.011	0.188 ± 0.013	0.226 ± 0.014	0.261 ± 0.015	0.277 ± 0.015	0.282 ± 0.015	0.283 ± 0.015 (k=59)	0.283 ± 0.015
annealed	exit (entrants)	0.401 ± 0.010	0.612 ± 0.009	0.756 ± 0.009	0.912 ± 0.005	0.992 ± 0.001	–	–	1.000 ± 0.000 (k=15)	1.000 ± 0.000
annealed	exit (orig. cohort)	0.116 ± 0.019	0.395 ± 0.022	0.635 ± 0.026	0.864 ± 0.021	0.989 ± 0.002	1.000 ± 0.000	–	1.000 ± 0.000 (k=18)	1.000 ± 0.000
quenched	re-entry	0.128 ± 0.016	0.177 ± 0.018	0.216 ± 0.024	0.257 ± 0.027	0.299 ± 0.028	0.320 ± 0.027	0.328 ± 0.026	0.330 ± 0.026 (k=59)	0.328 ± 0.026
quenched	exit (entrants)	0.388 ± 0.023	0.598 ± 0.037	0.740 ± 0.041	0.893 ± 0.026	0.983 ± 0.013	0.999 ± 0.001	0.998 ± nan	0.999 ± 0.000 (k=23)	0.992 ± 0.004
quenched	exit (orig. cohort)	0.116 ± 0.019	0.395 ± 0.019	0.651 ± 0.034	0.868 ± 0.026	0.978 ± 0.011	0.993 ± 0.004	0.997 ± 0.002	0.998 ± 0.001 (k=60)	0.998 ± 0.001
cosine	re-entry	0.085 ± 0.000	0.180 ± 0.000	0.299 ± 0.000	0.334 ± 0.000	0.345 ± 0.000	0.346 ± 0.000	0.352 ± 0.000	0.366 ± 0.000 (k=59)	0.359 ± 0.000
cosine	exit (entrants)	0.451 ± 0.000	0.821 ± 0.000	0.972 ± 0.000	0.998 ± 0.000	–	–	–	1.000 ± 0.000 (k=6)	0.993 ± 0.000
cosine	exit (orig. cohort)	0.123 ± 0.000	0.355 ± 0.000	0.489 ± 0.000	0.720 ± 0.000	0.875 ± 0.000	0.947 ± 0.000	0.974 ± 0.000	0.984 ± 0.000 (k=60)	0.984 ± 0.000

2C. Does "returned before" raise the re-entry hazard at fixed depth?

Depth = S at N−1 of the parcel at risk of re-entering. Counts are per seed. Mantel–Haenszel rate ratio over 7 depth bins × age.

mode	stratum	h(returned before) / h(first)	events/at risk returned (per seed)	events/at risk first (per seed)
annealed	all depths (crude)	1.16 ± 0.08		
annealed	S_{N-1} (2,2.5]	0.93 ± 0.05	77489/389408	200135/937870
annealed	S_{N-1} (2.5,3]	0.91 ± 0.05	17462/319078	45984/766635
annealed	S_{N-1} (3,5]	1.09 ± 0.06	6693/867197	15411/2168631
annealed	S_{N-1} >5	1.33 ± 0.65	40/11293294	105/34455341
annealed	MH, depth-adjusted	0.94 ± 0.04		
annealed	MH, age-adjusted	0.98 ± 0.06		
annealed	MH, depth × age	0.94 ± 0.04		
quenched	all depths (crude)	1.77 ± 0.40		
quenched	S_{N-1} (2,2.5]	1.01 ± 0.15	126974/578442	212890/962966
quenched	S_{N-1} (2.5,3]	1.09 ± 0.25	31359/486948	48484/792871
quenched	S_{N-1} (3,5]	1.24 ± 0.33	11633/1314831	16892/2283813
quenched	S_{N-1} >5	2.47 ± 2.96	49/10787467	95/33479354
quenched	MH, depth-adjusted	1.03 ± 0.17		
quenched	MH, age-adjusted	1.34 ± 0.24		
quenched	MH, depth × age	1.03 ± 0.15		
cosine	all depths (crude)	2.57 ± 0.00		
cosine	S_{N-1} (2,2.5]	0.91 ± 0.00	191249/853890	242045/986164
cosine	S_{N-1} (2.5,3]	0.96 ± 0.00	29296/856907	23682/665021
cosine	S_{N-1} (3,5]	0.43 ± 0.00	3992/2524131	9492/2568860
cosine	S_{N-1} >5	nan ± nan	0/7479489	0/32676324
cosine	MH, depth-adjusted	0.90 ± 0.00		
cosine	MH, age-adjusted	1.60 ± 0.00		
cosine	MH, depth × age	1.09 ± 0.00		

Distribution of S

4C. Kaplan (PRL 89, 184103) log-normal tail test, fit N = 20..60

α, v = slopes of mean and variance of ln S on the Eulerian window, N = 20..60; measured = slope of −ln P(S ≤ s) over the same N (points with ≥ 10 parcels). Ratio = measured / predicted.

mode	alpha	v	pred a²/2v	s=1.25: rate (ratio)	s=1.5: rate (ratio)	s=2: rate (ratio)	s=3: rate (ratio)	s=5: rate (ratio)	s=10: rate (ratio)
annealed	0.1526 ± 0.0022	0.1623 ± 0.0028	0.0717 ± 0.0014	0.1096 ± 0.0030 (1.53)	0.1093 ± 0.0021 (1.52)	0.1085 ± 0.0018 (1.51)	0.1068 ± 0.0018 (1.49)	0.1043 ± 0.0018 (1.46)	0.0996 ± 0.0016 (1.39)
quenched	0.1452 ± 0.0094	0.2891 ± 0.0378	0.0370 ± 0.0061	0.0307 ± 0.0062 (0.85)	0.0316 ± 0.0066 (0.87)	0.0332 ± 0.0070 (0.92)	0.0348 ± 0.0074 (0.96)	0.0355 ± 0.0075 (0.97)	0.0381 ± 0.0073 (1.04)
cosine	0.0236 ± 0.0000	0.0223 ± 0.0000	0.0125 ± 0.0000	0.0211 ± 0.0000 (1.69)	0.0218 ± 0.0000 (1.75)	0.0210 ± 0.0000 (1.68)	0.0195 ± 0.0000 (1.56)	0.0183 ± 0.0000 (1.46)	0.0206 ± 0.0000 (1.65)

4C reproduction, seed 0

mode (seed 0)	predicted	measured s=2	measured s=1.25..10
annealed	0.0724	0.1088	0.0993–0.1118
quenched	0.0319	0.0266	0.0245–0.0277
cosine	0.0125	0.0210	0.0183–0.0218

4A. P(S ≤ s) (mean over seeds)

mode	N	s=1.25	s=1.5	s=2	s=3	s=5	s=10	s=20	s=50	s=100	s=1000	s=10000	s=1e+06
annealed	10	7.53e-03	0.0249	0.0723	0.1756	0.3489	0.6015	0.7991	0.9420	0.9839	1.0000	1.0000	1.0000
annealed	20	1.54e-03	5.16e-03	0.0155	0.0402	0.0924	0.2032	0.3521	0.5710	0.7208	0.9695	0.9992	1.0000
annealed	40	1.41e-04	4.79e-04	1.47e-03	4.00e-03	9.86e-03	0.0252	0.0529	0.1161	0.1872	0.5334	0.8430	0.9973
annealed	60	1.84e-05	6.65e-05	1.91e-04	5.40e-04	1.39e-03	3.75e-03	8.48e-03	0.0210	0.0380	0.1698	0.4300	0.9056
quenched	10	0.0104	0.0320	0.0839	0.1875	0.3573	0.6136	0.8185	0.9524	0.9880	0.9999	1.0000	1.0000
quenched	20	3.73e-03	0.0116	0.0293	0.0636	0.1247	0.2406	0.3831	0.5922	0.7366	0.9765	0.9995	1.0000
quenched	40	1.69e-03	4.69e-03	0.0116	0.0247	0.0452	0.0810	0.1247	0.1917	0.2544	0.5482	0.8369	0.9978
quenched	60	1.01e-03	3.05e-03	7.54e-03	0.0153	0.0285	0.0507	0.0748	0.1059	0.1302	0.2463	0.4491	0.8874
cosine	10	0.0143	0.0573	0.1490	0.2467	0.4236	0.8011	0.9389	0.9731	0.9861	0.9999	1.0000	1.0000
cosine	20	4.43e-03	0.0169	0.0621	0.1304	0.2194	0.4652	0.7843	0.9511	0.9760	0.9990	0.9999	1.0000
cosine	40	6.20e-03	0.0268	0.0458	0.0716	0.1201	0.2439	0.4821	0.8563	0.9441	0.9926	0.9971	0.9999
cosine	60	2.59e-03	0.0122	0.0257	0.0497	0.1021	0.1916	0.3475	0.7481	0.8988	0.9868	0.9934	0.9988

4A. Moments and quantiles of ln S

mode	N	mean ln S	var ln S	skew	ex. kurt	q1	q10	q25	q50	q75	q90	q99
annealed	10	2.116 ± 0.045	1.088 ± 0.022	0.483 ± 0.053	-0.113 ± 0.096	0.26	0.82	1.33	2.02	2.79	3.55	4.81
annealed	30	5.236 ± 0.062	4.126 ± 0.069	0.252 ± 0.017	-0.099 ± 0.024	1.01	2.65	3.80	5.15	6.58	7.91	10.26
annealed	60	9.807 ± 0.092	9.041 ± 0.112	0.132 ± 0.008	-0.057 ± 0.011	3.16	5.97	7.73	9.74	11.81	13.71	17.05
quenched	10	2.063 ± 0.085	1.039 ± 0.050	0.453 ± 0.125	-0.003 ± 0.230	0.23	0.77	1.30	1.99	2.72	3.43	4.68
quenched	30	5.014 ± 0.268	4.800 ± 0.318	0.094 ± 0.152	-0.346 ± 0.149	0.52	2.05	3.45	5.03	6.54	7.85	10.07
quenched	60	9.366 ± 0.549	14.182 ± 1.580	-0.265 ± 0.158	-0.292 ± 0.230	0.89	3.81	7.00	9.71	12.01	13.95	17.19
cosine	10	1.726 ± 0.000	0.844 ± 0.000	1.034 ± 0.000	2.976 ± 0.000	0.19	0.51	1.11	1.75	2.18	2.70	4.91
cosine	30	2.696 ± 0.000	1.247 ± 0.000	0.732 ± 0.000	5.074 ± 0.000	0.20	1.26	2.11	2.75	3.28	3.91	5.94
cosine	60	3.317 ± 0.000	1.944 ± 0.000	1.957 ± 0.000	15.366 ± 0.000	0.36	1.59	2.62	3.39	3.92	4.61	7.72

4D. i.i.d. random-matrix model vs the 2-D annealed run

I(0+) from the density of λ = ln S/N at its lowest bin, difference estimator between N = 40 and 60 (removes the prefactor); L(q) the same way.

model	alpha	v	Gaussian $a^2/2v$	decay of P(S≤2) = I(0)	decay s=1.25..10	λ-CDF I(0+), N 40→60 → corrected for (ln S)²	−min L(q): N 20→40 / 40→60
i.i.d. model (1e7)	0.1543	0.1629	0.0731	0.1090	0.1011–0.1103	0.0702 → 0.1107	0.1181 / 0.1011
2-D annealed (10 seeds)	0.1526 ± 0.0022	0.1623 ± 0.0028	0.0717 ± 0.0014	0.1085 ± 0.0018	0.0996–0.1096		

4D. P(S ≤ s, N): i.i.d. vs 2-D annealed (z = difference in standard errors of the seed mean)

N	iid s=1.5	2-D s=1.5	z	iid s=2	2-D s=2	z	iid s=5	2-D s=5	z	iid s=10	2-D s=10	z
5	7.971e-02	8.260e-02 ± 1.1e-02	-0.8	2.129e-01	2.196e-01 ± 2.6e-02	-0.8	7.064e-01	7.146e-01 ± 1.9e-02	-1.3	9.240e-01	9.218e-01 ± 2.4e-03	+2.8
10	2.464e-02	2.486e-02 ± 3.3e-03	-0.2	7.153e-02	7.234e-02 ± 8.3e-03	-0.3	3.485e-01	3.489e-01 ± 1.9e-02	-0.1	6.004e-01	6.015e-01 ± 1.9e-02	-0.2
20	4.950e-03	5.162e-03 ± 6.7e-04	-1.0	1.491e-02	1.546e-02 ± 1.9e-03	-0.9	9.114e-02	9.239e-02 ± 8.3e-03	-0.5	2.015e-01	2.032e-01 ± 1.2e-02	-0.4
40	4.440e-04	4.794e-04 ± 5.0e-05	-2.2	1.368e-03	1.466e-03 ± 1.5e-04	-2.1	9.503e-03	9.863e-03 ± 7.1e-04	-1.6	2.439e-02	2.516e-02 ± 1.6e-03	-1.5
60	6.030e-05	6.650e-05 ± 9.8e-06	-2.0	1.869e-04	1.914e-04 ± 2.3e-05	-0.6	1.287e-03	1.388e-03 ± 1.3e-04	-2.4	3.497e-03	3.753e-03 ± 3.6e-04	-2.3

4D. Generalised Lyapunov exponents L(q)

For q ≥ 0 the transfer operator on 2048 directions × 301 fp bins gives the exact vector-norm exponent (= σmax exponent). For q < 0, σmax ≥ 1 makes L(q) plateau at −I(0); the Monte-Carlo difference estimator shows it. Large positive q is dominated by rare samples in Monte Carlo.

q	L(q) transfer operator (vector, exact)	L(q) MC, N 40→60	L(q) MC, (1/60) ln E
0.00	-0.0000	0.0000	-0.0000
0.25	0.0438	0.0438	0.0460
0.50	0.0984	0.0984	0.1022
0.75	0.1641	0.1642	0.1690
1.00	0.2411	0.2416	0.2466
1.25	0.3290	0.3291	0.3342
1.50	0.4271	0.4219	0.4296
1.75	0.5344	0.5141	0.5300
2.00	0.6497	0.6024	0.6332
2.25	0.7719	0.6862	0.7381
2.50	0.9002	0.7663	0.8438
2.75	1.0336	0.8437	0.9501
3.00	1.1717	0.9193	1.0569
3.25	1.3139	0.9939	1.1639
3.50	1.4599	1.0677	1.2711
3.75	1.6094	1.1411	1.3784
4.00	1.7621	1.2141	1.4859

4E. Robustness: threshold and metric

S_f = σmax(D M D^{-1}), D = diag(1, f): the same dynamics measured with y stretched by f.

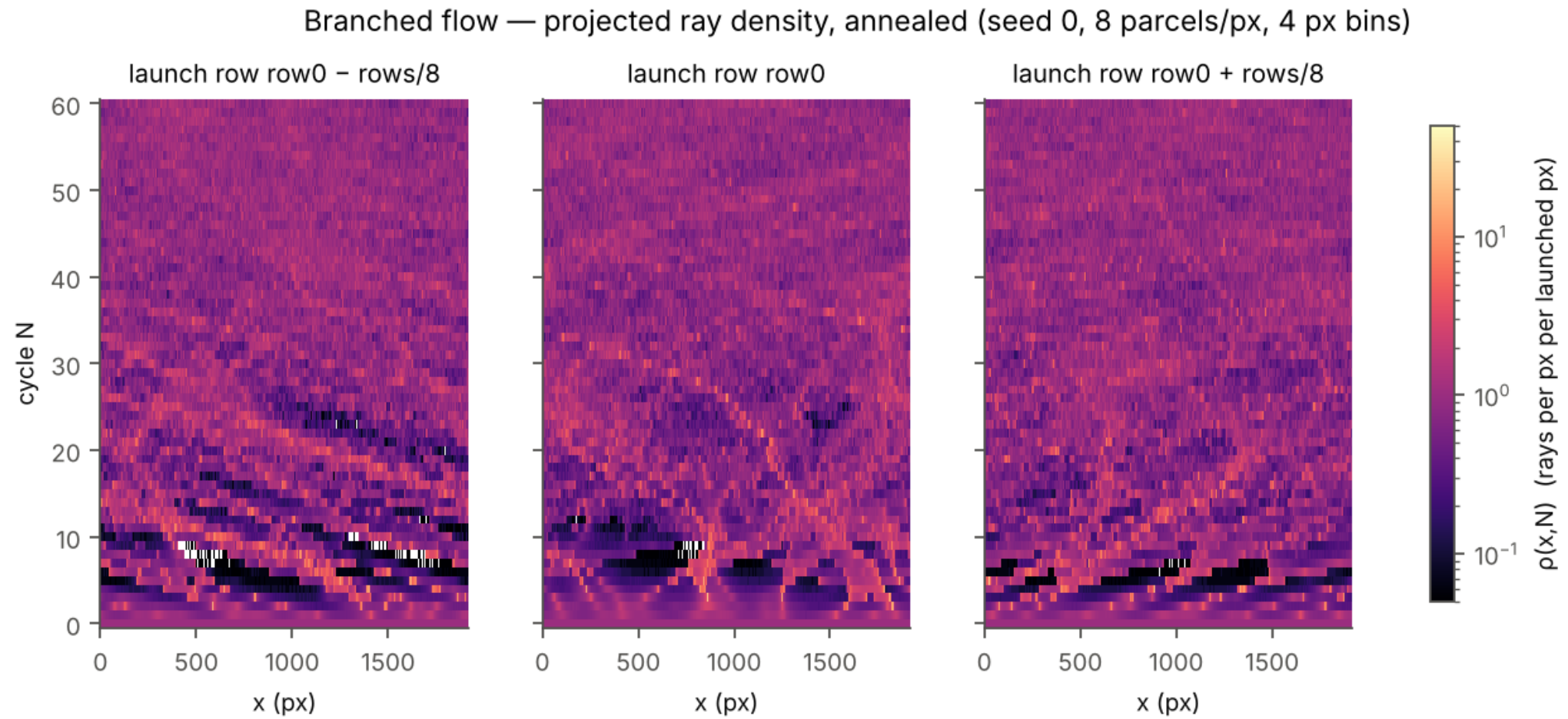
mode	y scale f	alpha	rate S_THR=1.5	P(N=60)	rate S_THR=2	P(N=60)	rate S_THR=3	P(N=60)
annealed	1	0.1526 ± 0.0022	0.1093 ± 0.0021	6.65e-05	0.1085 ± 0.0018	1.91e-04	0.1068 ± 0.0018	5.40e-04
annealed	0.5	0.1525 ± 0.0022	0.1075 ± 0.0020	4.97e-05	0.1066 ± 0.0018	1.66e-04	0.1054 ± 0.0017	4.67e-04
annealed	2	0.1526 ± 0.0023	0.1086 ± 0.0023	6.19e-05	0.1079 ± 0.0022	1.87e-04	0.1065 ± 0.0021	5.28e-04
annealed	4	0.1525 ± 0.0023	0.1060 ± 0.0023	5.00e-05	0.1055 ± 0.0019	1.53e-04	0.1046 ± 0.0020	4.31e-04
quenched	1	0.1452 ± 0.0094	0.0316 ± 0.0066	3.05e-03	0.0332 ± 0.0070	7.54e-03	0.0348 ± 0.0074	1.53e-02
quenched	0.5	0.1451 ± 0.0094	0.0308 ± 0.0066	2.25e-03	0.0317 ± 0.0065	6.19e-03	0.0332 ± 0.0067	1.33e-02
quenched	2	0.1451 ± 0.0094	0.0355 ± 0.0074	2.08e-03	0.0356 ± 0.0078	5.68e-03	0.0361 ± 0.0076	1.34e-02
quenched	4	0.1451 ± 0.0094	0.0361 ± 0.0076	1.26e-03	0.0373 ± 0.0081	3.45e-03	0.0381 ± 0.0083	8.68e-03
cosine	1	0.0236 ± 0.0000	0.0218 ± 0.0000	1.22e-02	0.0210 ± 0.0000	2.57e-02	0.0195 ± 0.0000	4.97e-02
cosine	0.5	0.0234 ± 0.0000	0.0231 ± 0.0000	1.17e-02	0.0225 ± 0.0000	2.43e-02	0.0219 ± 0.0000	4.55e-02
cosine	2	0.0236 ± 0.0000	0.0127 ± 0.0000	3.21e-03	0.0142 ± 0.0000	9.93e-03	0.0163 ± 0.0000	3.74e-02
cosine	4	0.0235 ± 0.0000	0.0122 ± 0.0000	2.74e-03	0.0137 ± 0.0000	6.87e-03	0.0174 ± 0.0000	1.78e-02

4F. Rarefaction r vs ln S on the lines

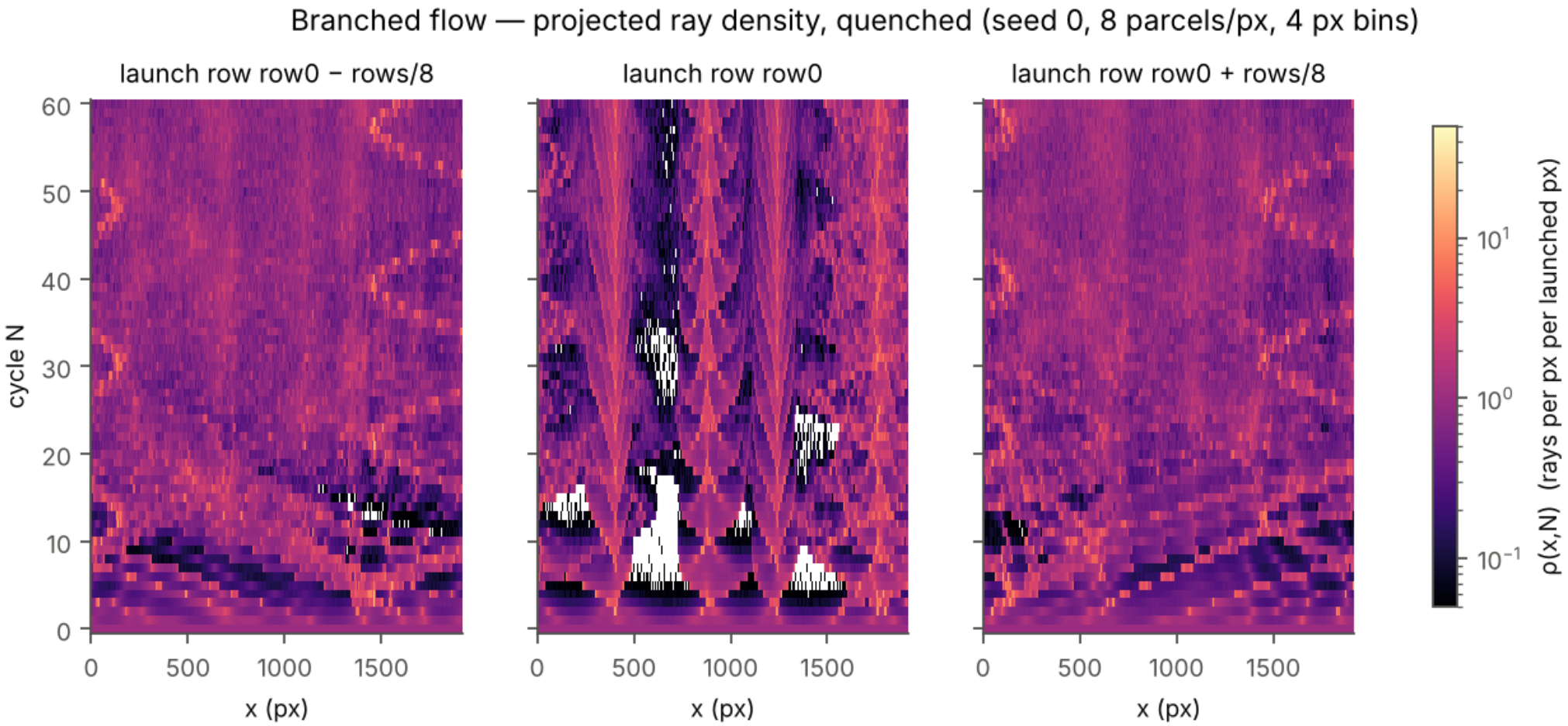
Pooled over the three lines. S ≥ e^r always, so P(r ≤ ln 2 | S ≤ 2) = 1 identically.

mode	N	P(S≤2)	P(S≤2 | r≤ln2)	P(r≤ln2 | S≤2)	r 10% quantile	P(S≤2 | r≤q10)	P(r≤q10 | S≤2)
annealed	10	0.0841 ± 0.0341	0.289 ± 0.072	1.000 ± 0.000	-0.07 ± 0.13	0.236 ± 0.066	0.301 ± 0.089
annealed	30	0.0042 ± 0.0017	0.110 ± 0.025	1.000 ± 0.000	1.76 ± 0.22	0.042 ± 0.017	1.000 ± 0.000
annealed	60	0.0002 ± 0.0002	0.062 ± 0.044	1.000 ± 0.000	5.10 ± 0.22	0.002 ± 0.002	1.000 ± 0.000
quenched	10	0.1375 ± 0.0397	0.389 ± 0.075	1.000 ± 0.000	-0.15 ± 0.07	0.211 ± 0.054	0.156 ± 0.018
quenched	30	0.0510 ± 0.0159	0.495 ± 0.073	1.000 ± 0.000	0.72 ± 0.32	0.452 ± 0.092	0.915 ± 0.143
quenched	60	0.0335 ± 0.0090	0.681 ± 0.101	1.000 ± 0.000	1.70 ± 0.64	0.333 ± 0.087	0.997 ± 0.009
cosine	10	0.3118 ± 0.0000	0.876 ± 0.000	1.000 ± 0.000	0.04 ± 0.00	0.881 ± 0.000	0.282 ± 0.000
cosine	30	0.1304 ± 0.0000	0.740 ± 0.000	1.000 ± 0.000	0.30 ± 0.00	0.870 ± 0.000	0.667 ± 0.000
cosine	60	0.0455 ± 0.0000	0.411 ± 0.000	1.000 ± 0.000	0.64 ± 0.00	0.386 ± 0.000	0.848 ± 0.000

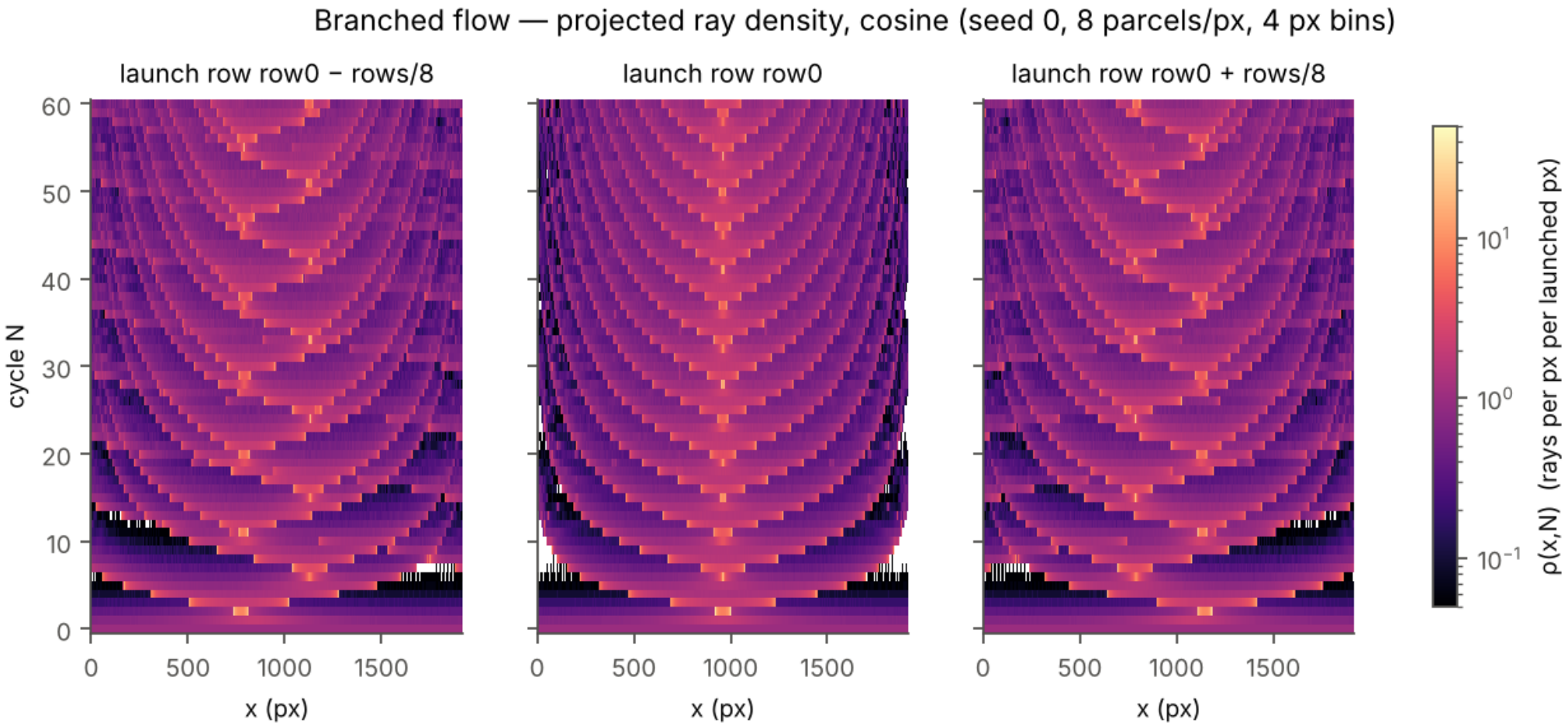
Figures



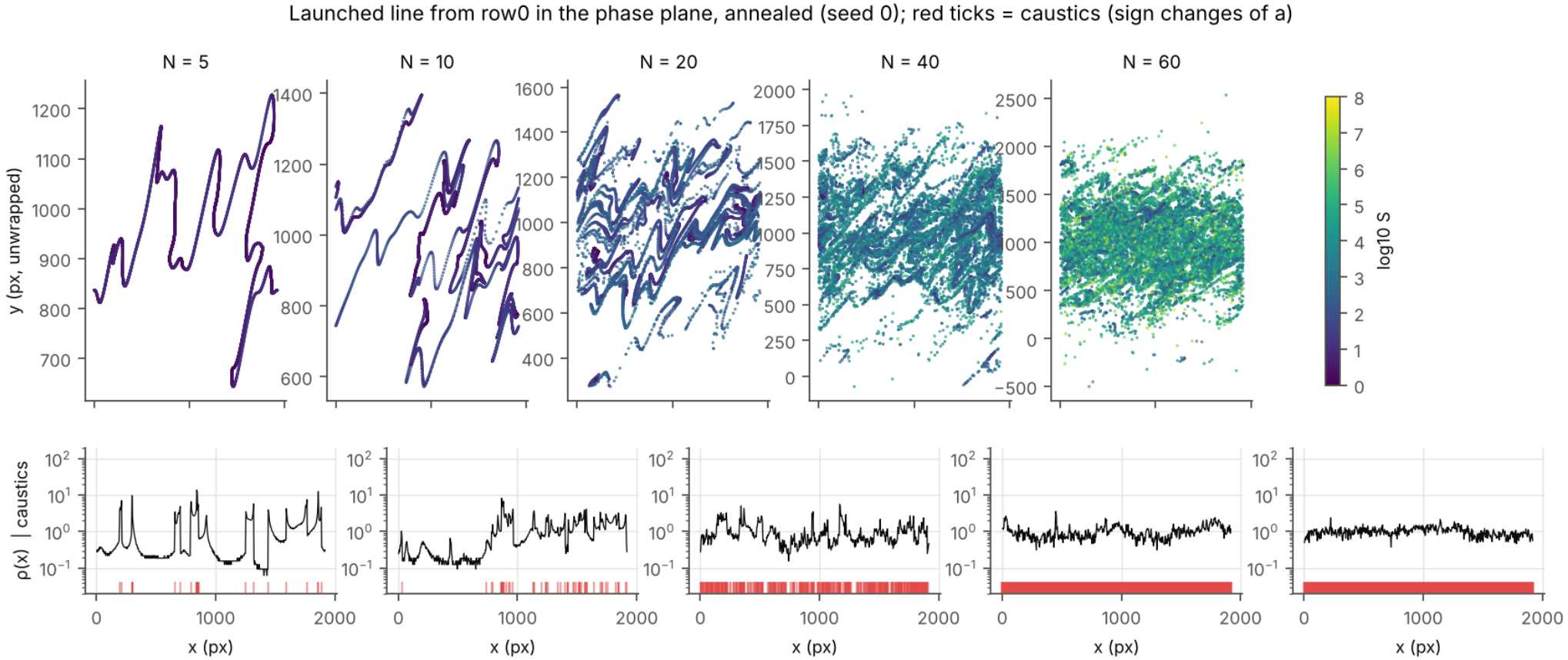
Branched-flow image ρ(x,N), annealed mode.



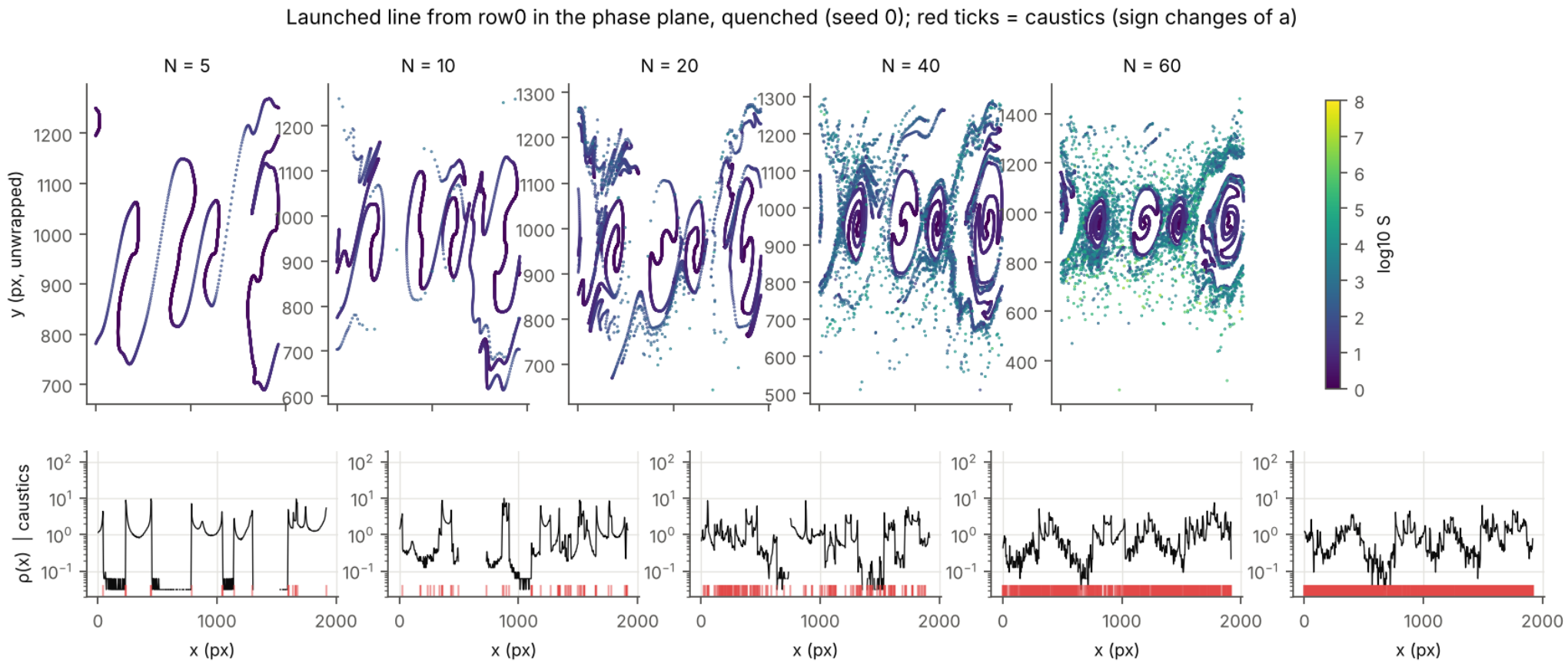
Branched-flow image ρ(x,N), quenched mode.



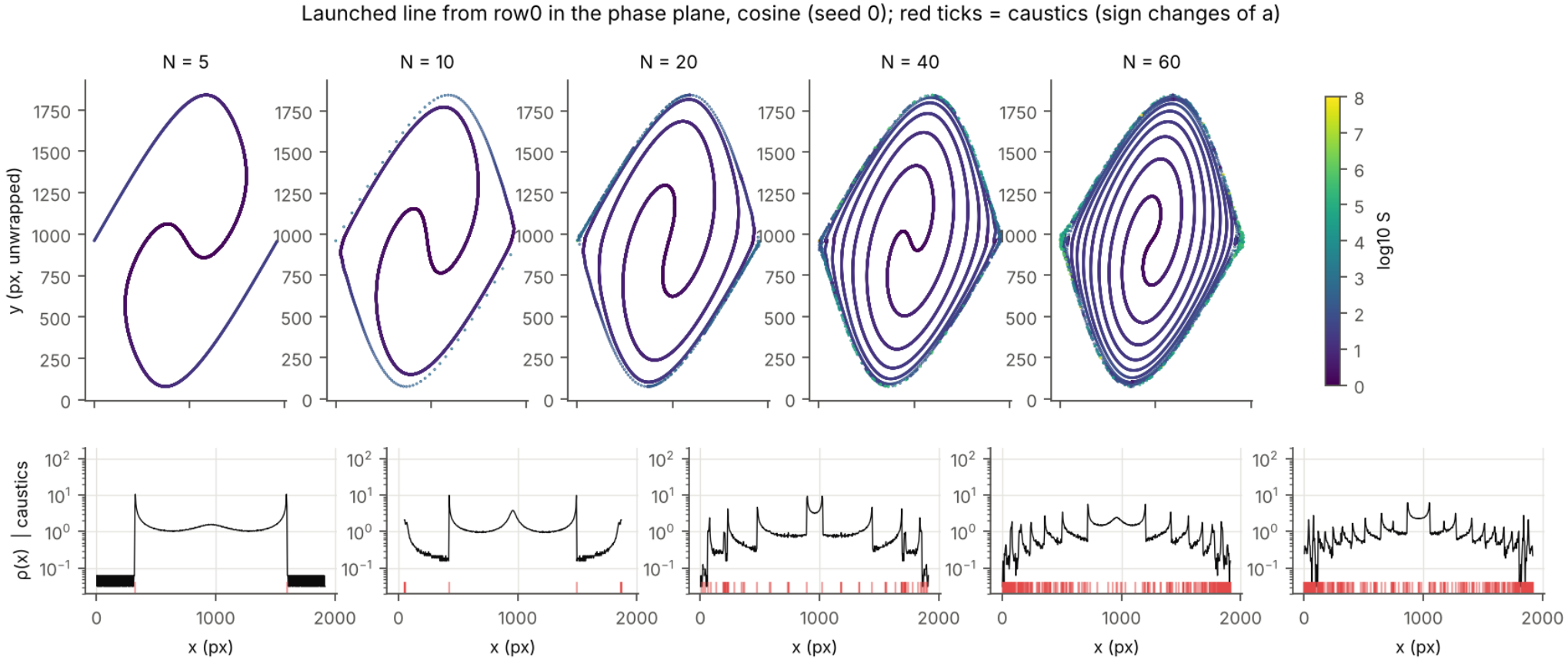
Branched-flow image ρ(x,N), cosine mode.



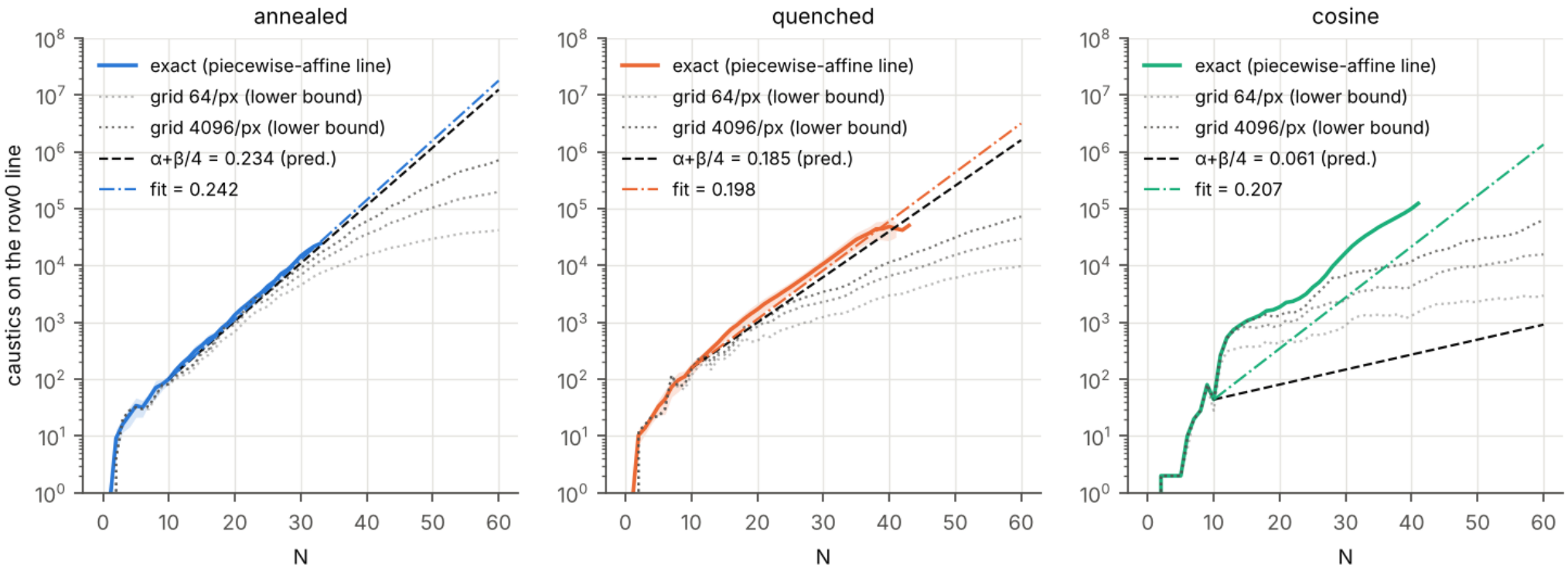
Manifold snapshots coloured by S with ρ(x) and caustics, annealed.



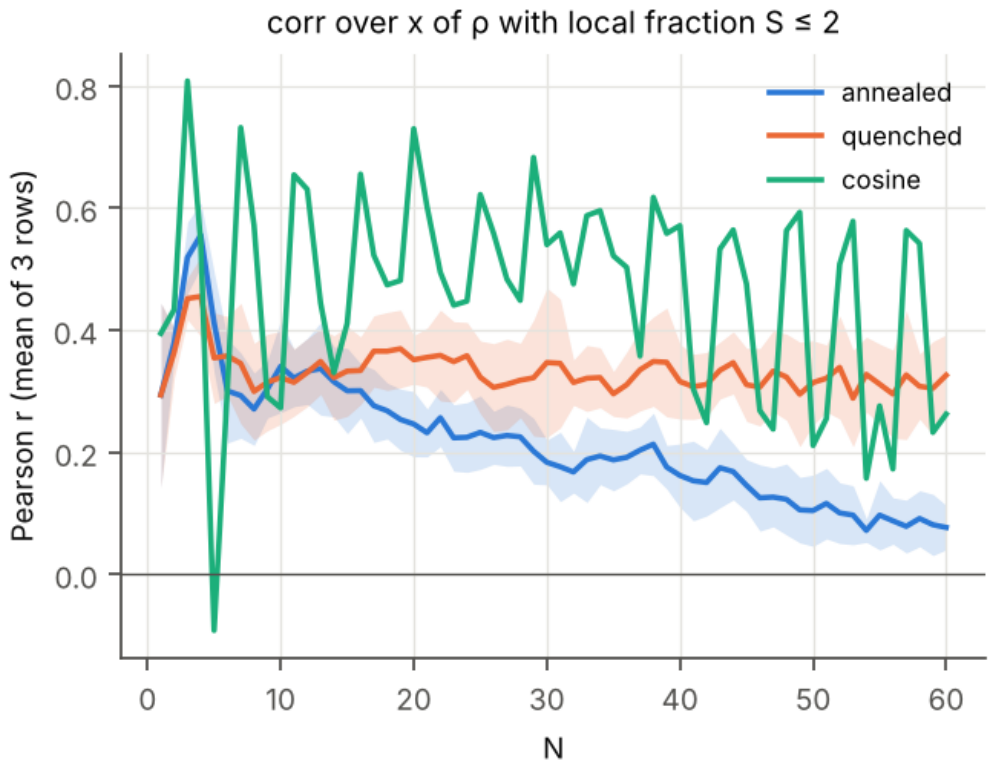
Manifold snapshots coloured by S with ρ(x) and caustics, quenched.

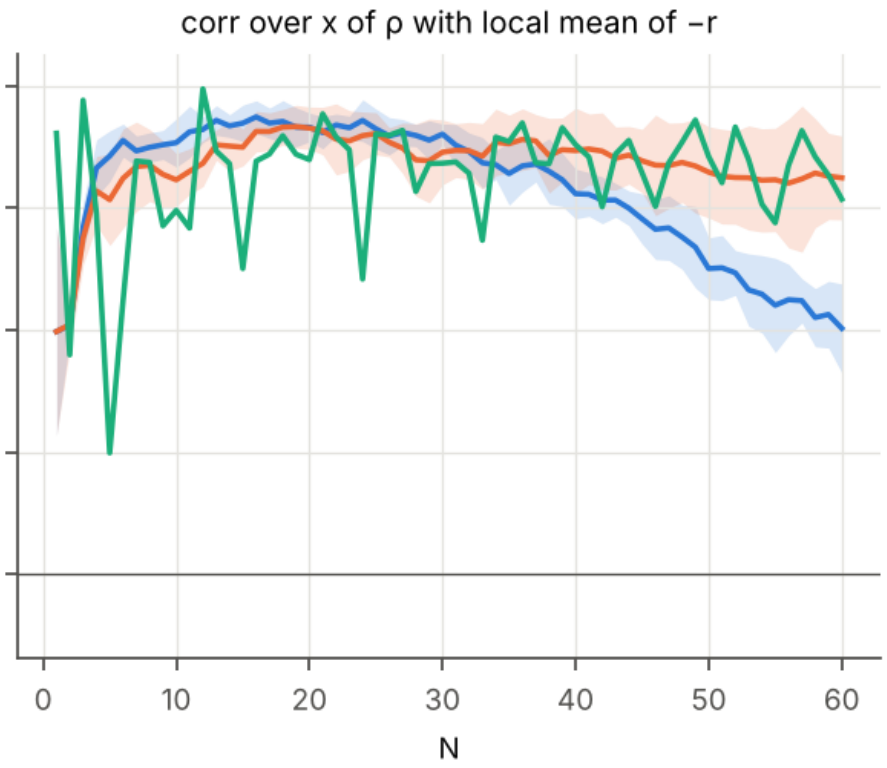


Manifold snapshots coloured by S with ρ(x) and caustics, cosine.

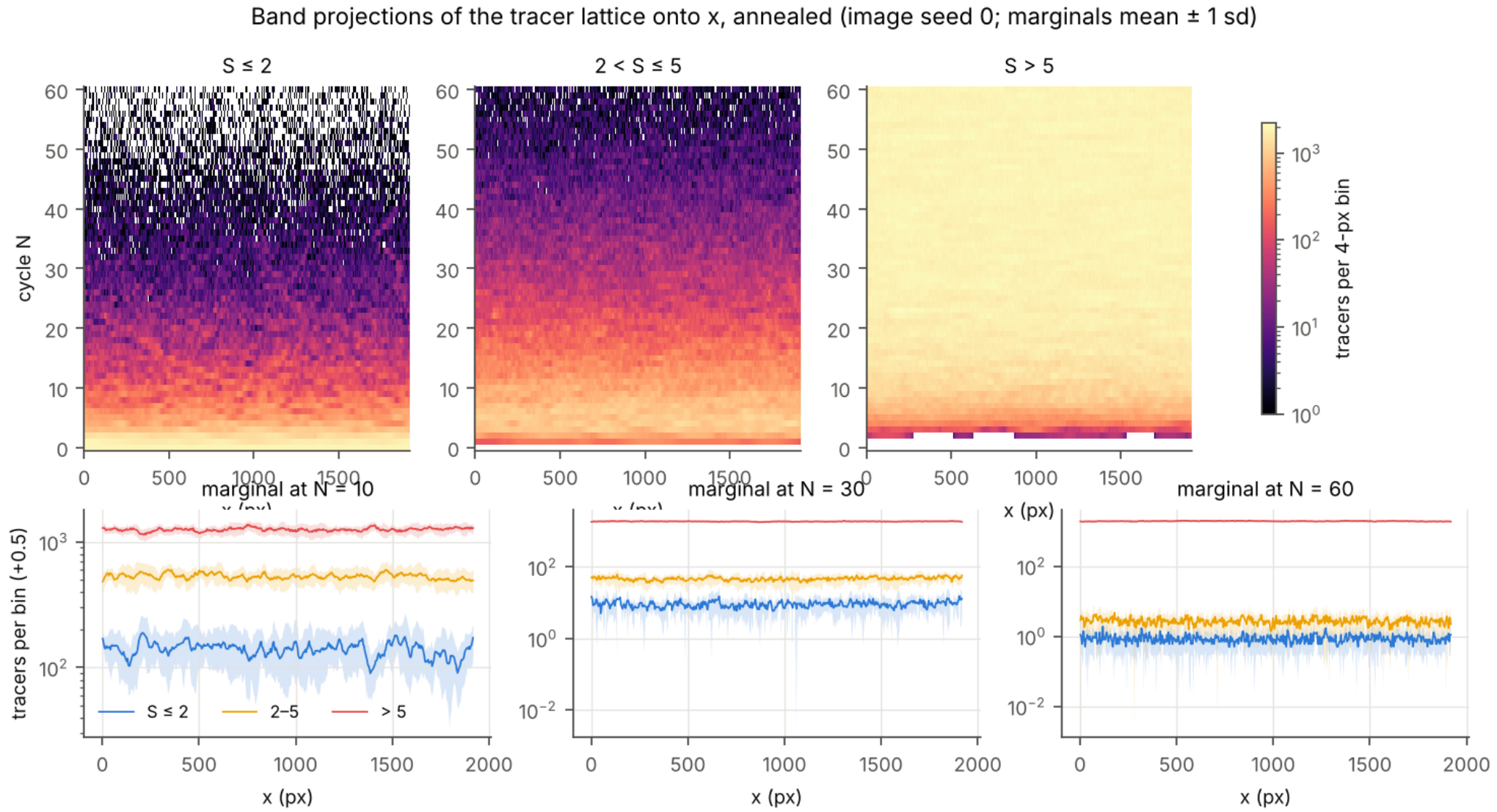


Caustic count vs N (exact, mean ± 1 sd over seeds) against Kaplan Eq. (6) rate α+β/4.

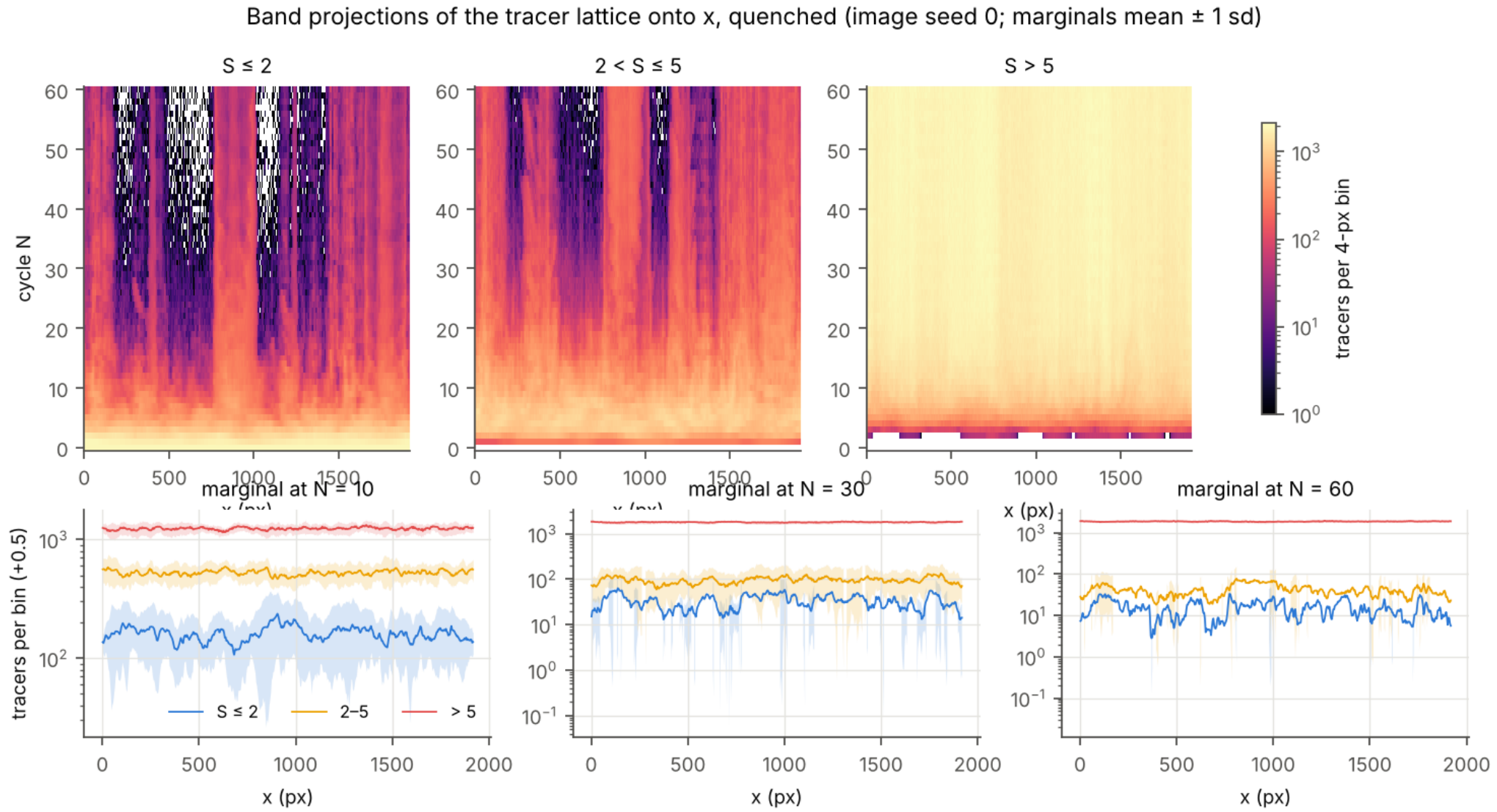




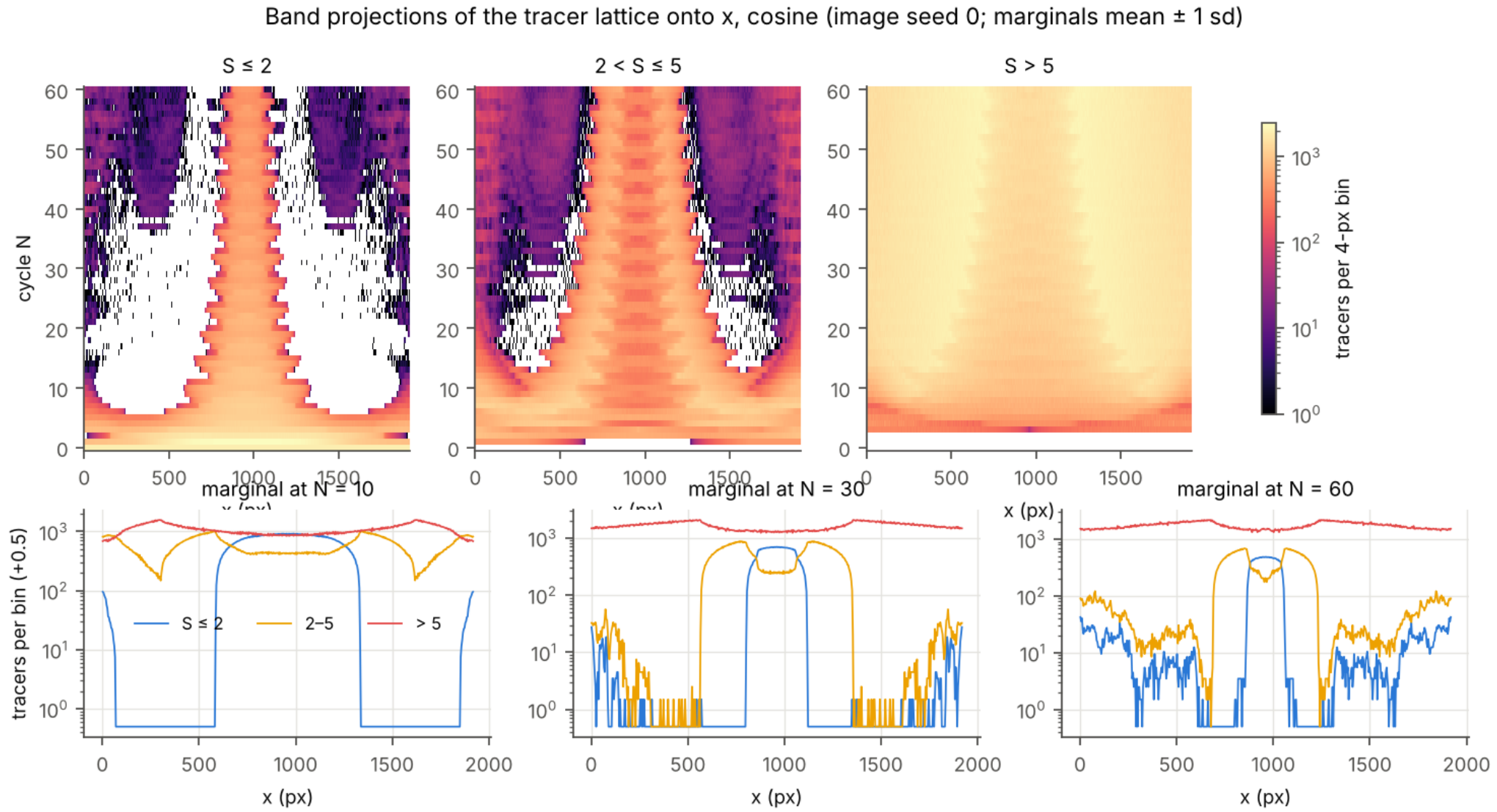
Correlation across x of ρ with low-stretch fraction and with −r.



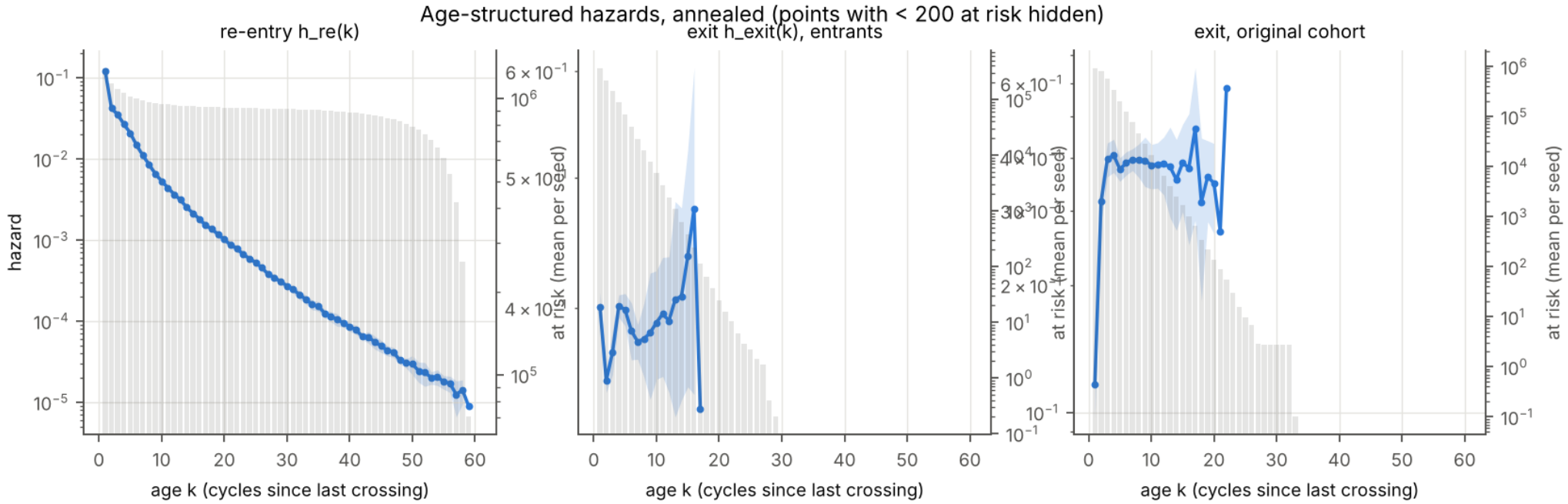
Band projections, annealed.



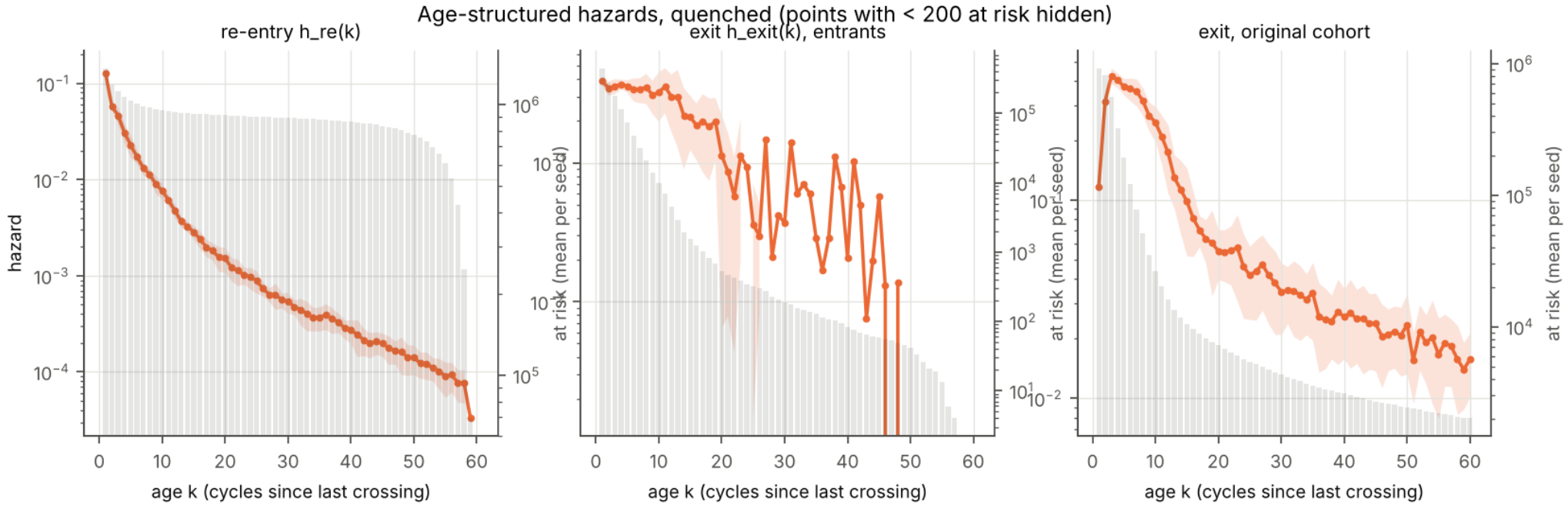
Band projections, quenched.



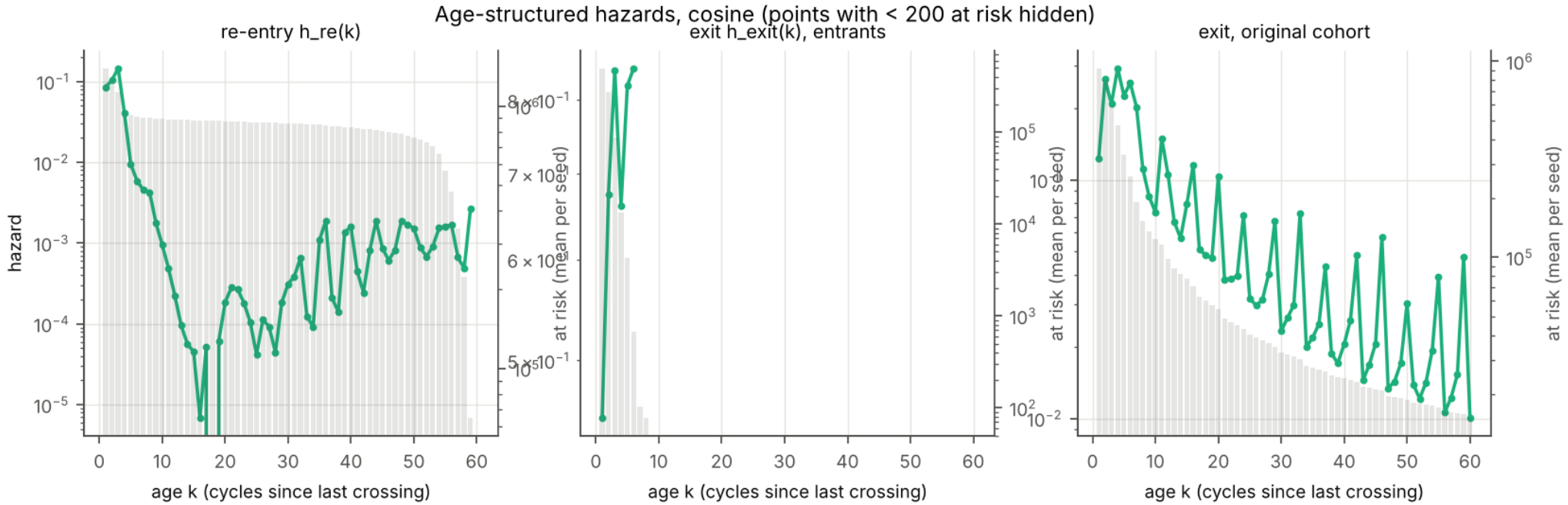
Band projections, cosine.



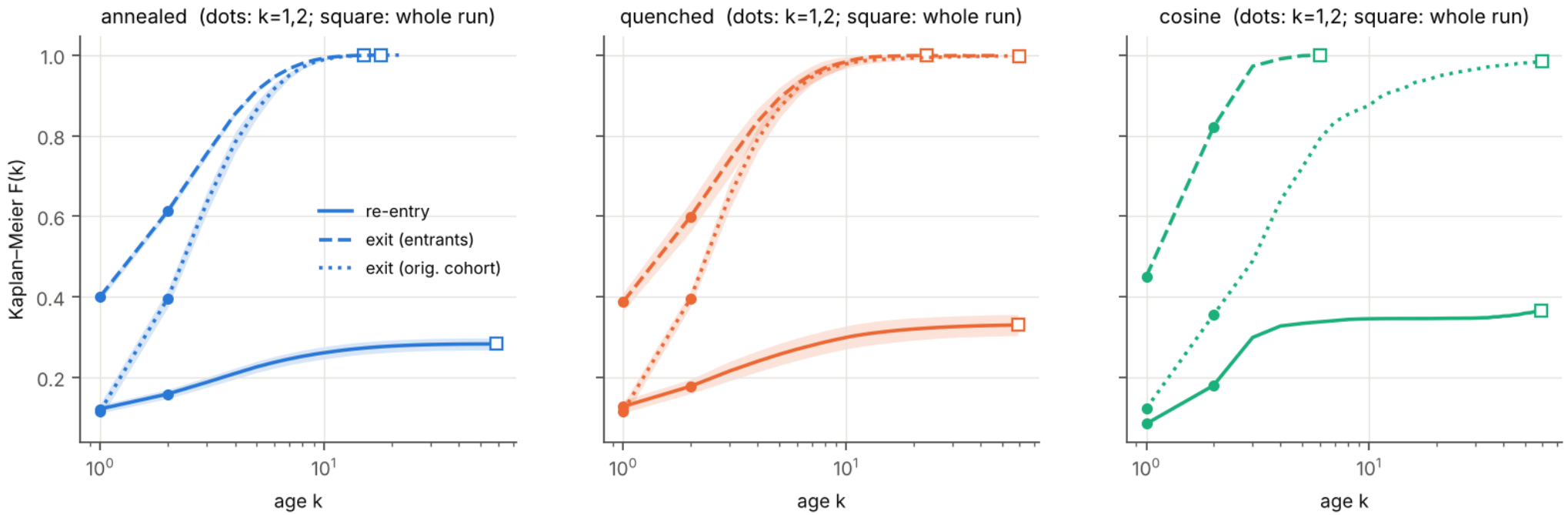
Hazards vs age with at-risk counts, annealed.



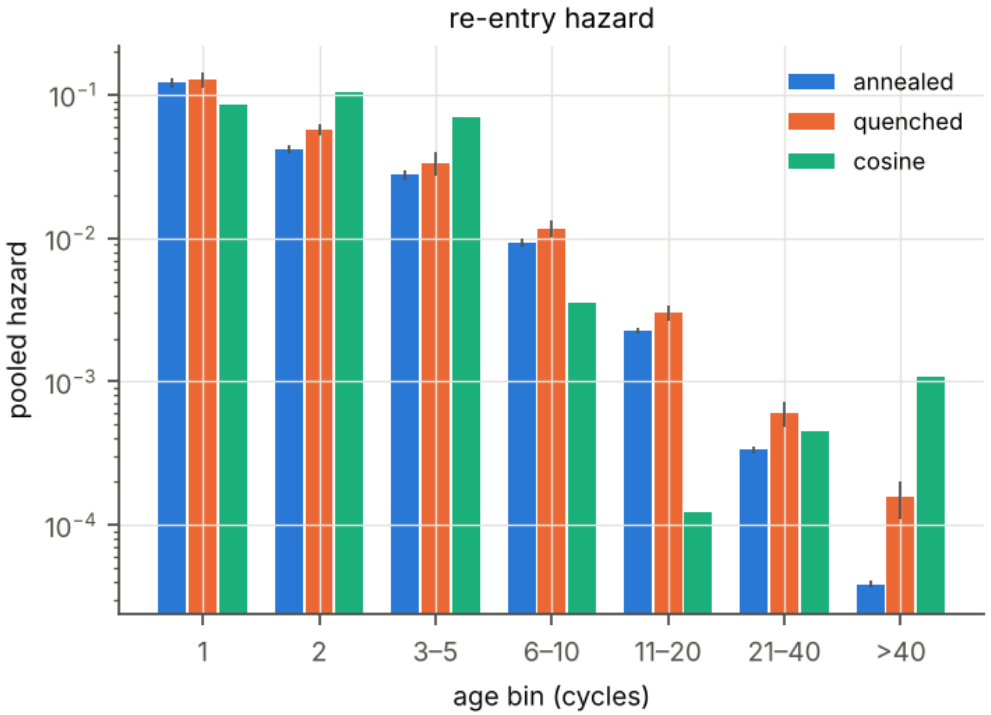
Hazards vs age with at-risk counts, quenched.

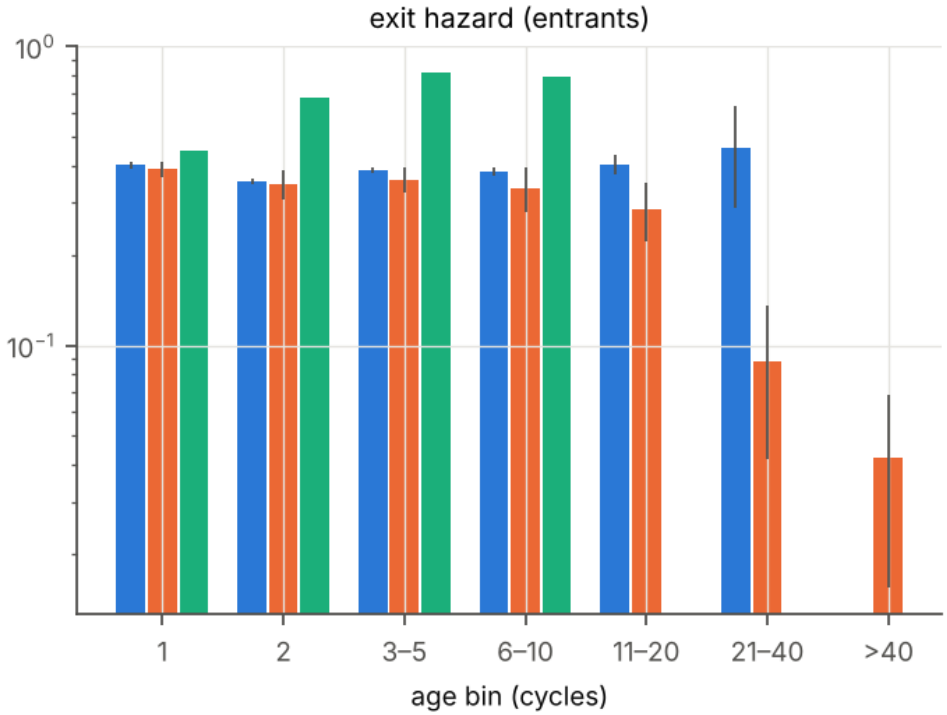


Hazards vs age with at-risk counts, cosine.

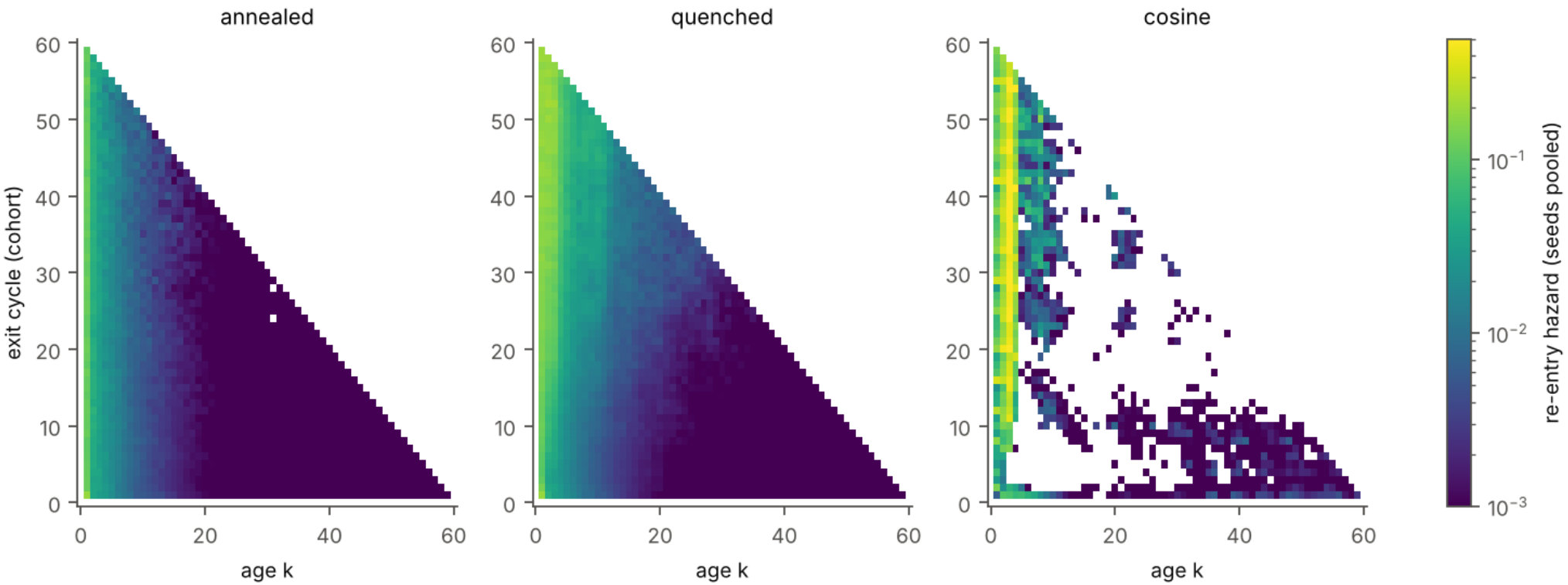


Kaplan–Meier cumulative probability of re-entry and exit.

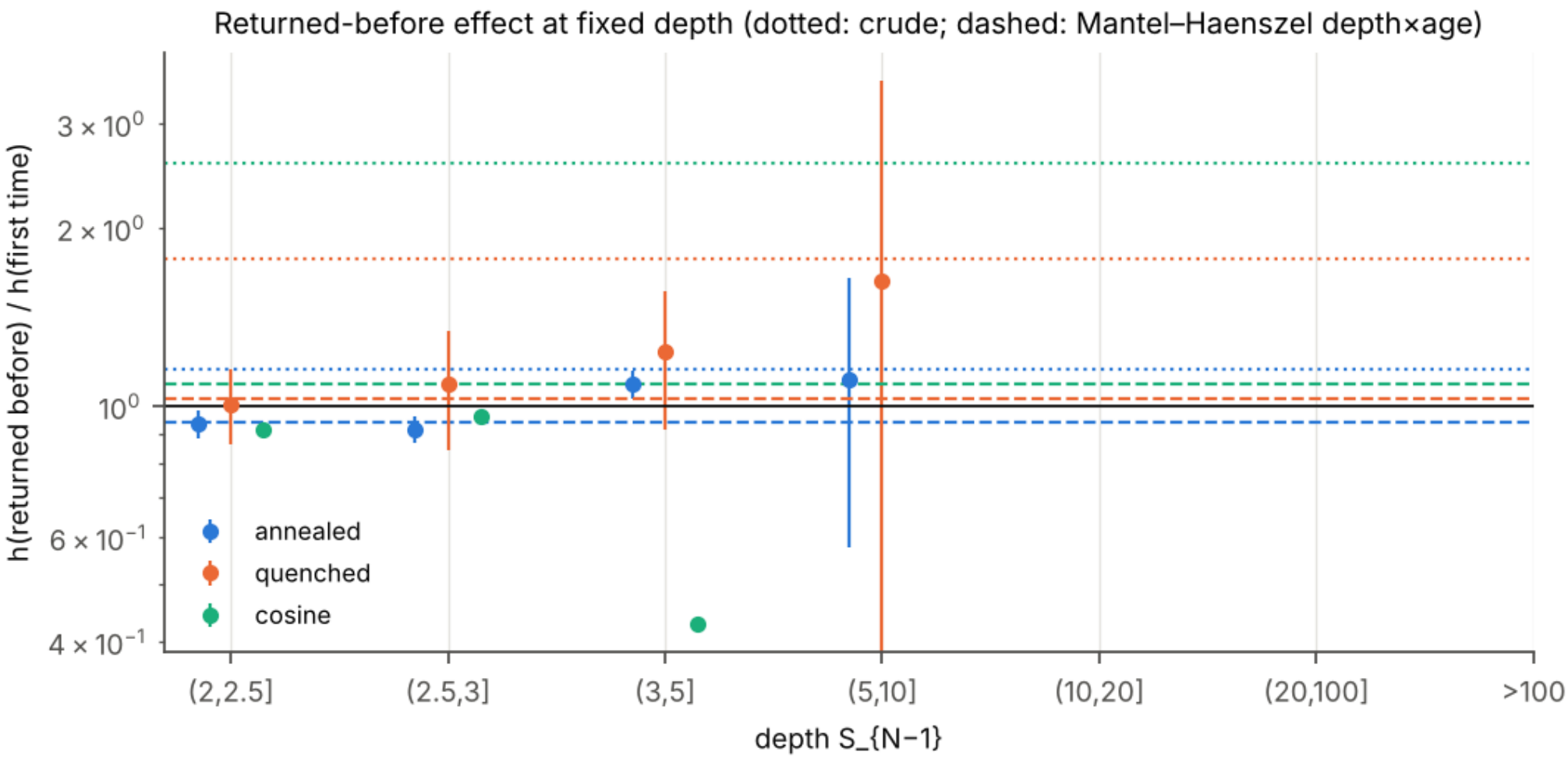




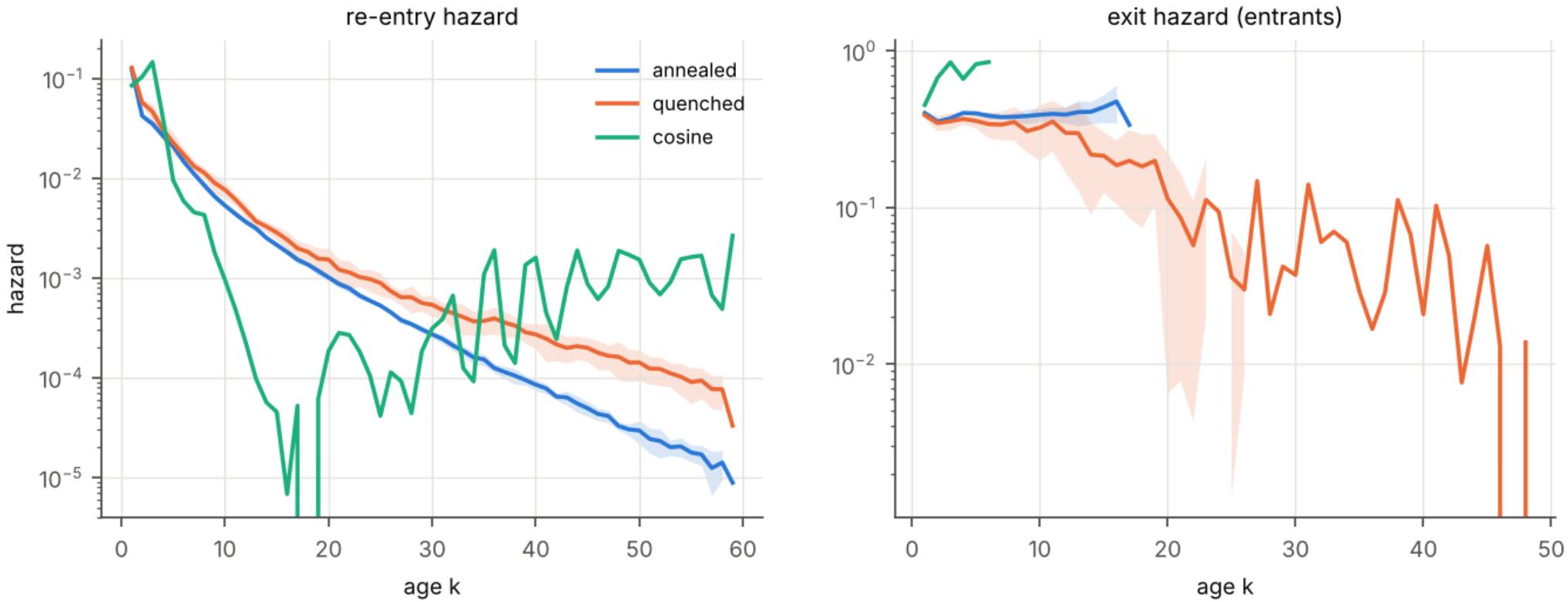
Pooled age-bin hazards, both directions, three modes.



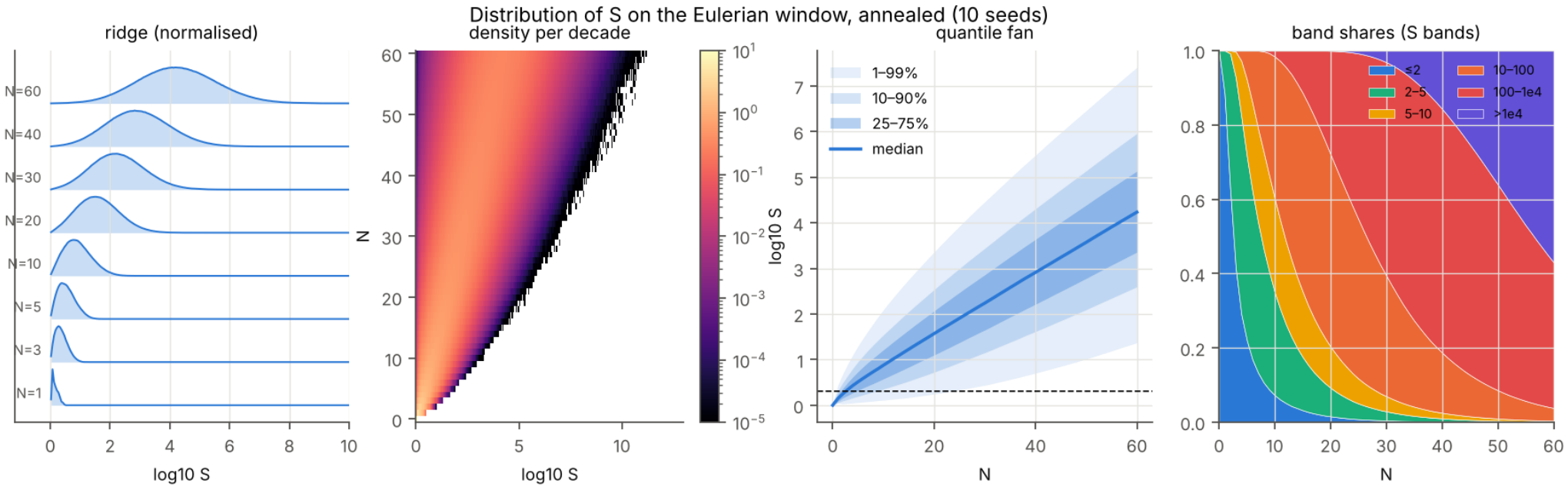
Cohort heat map: re-entry hazard by exit cycle and age.



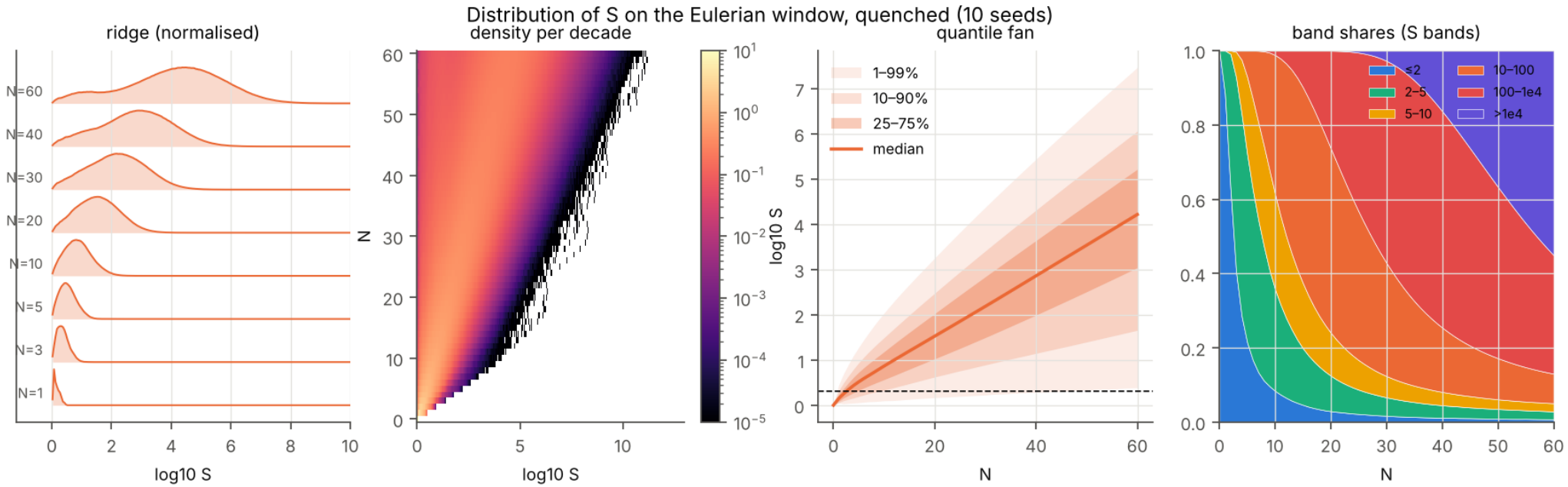
Depth-controlled re-entry ratio.



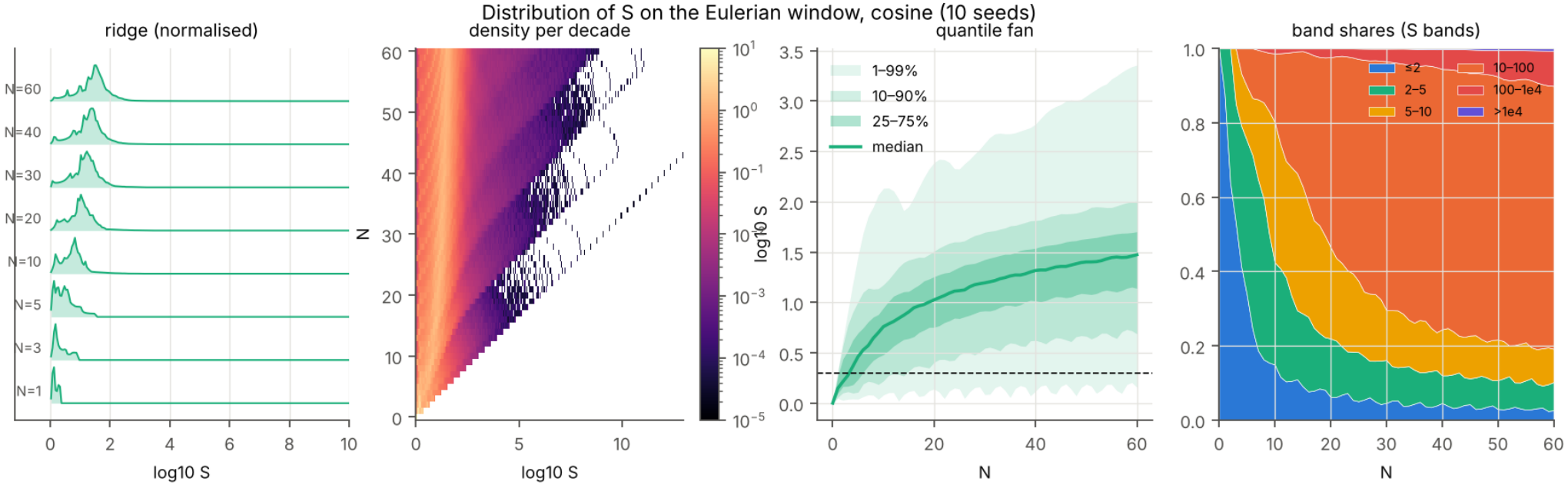
Three modes overlaid: hazards vs age.



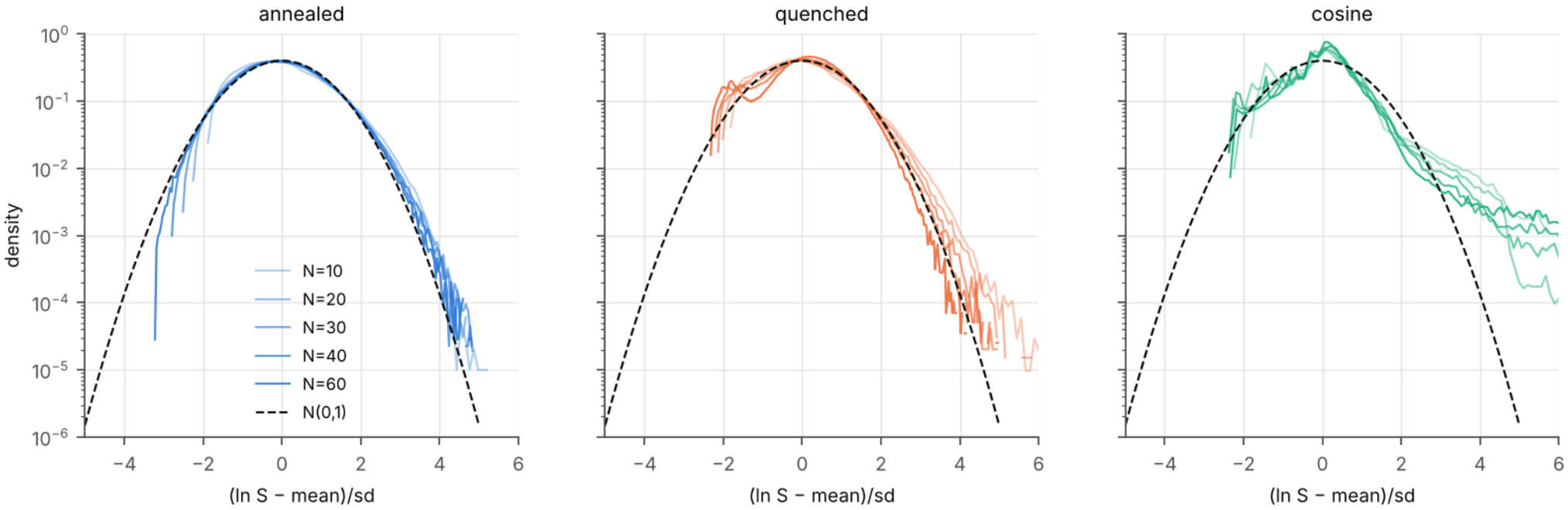
Distribution of S, annealed.



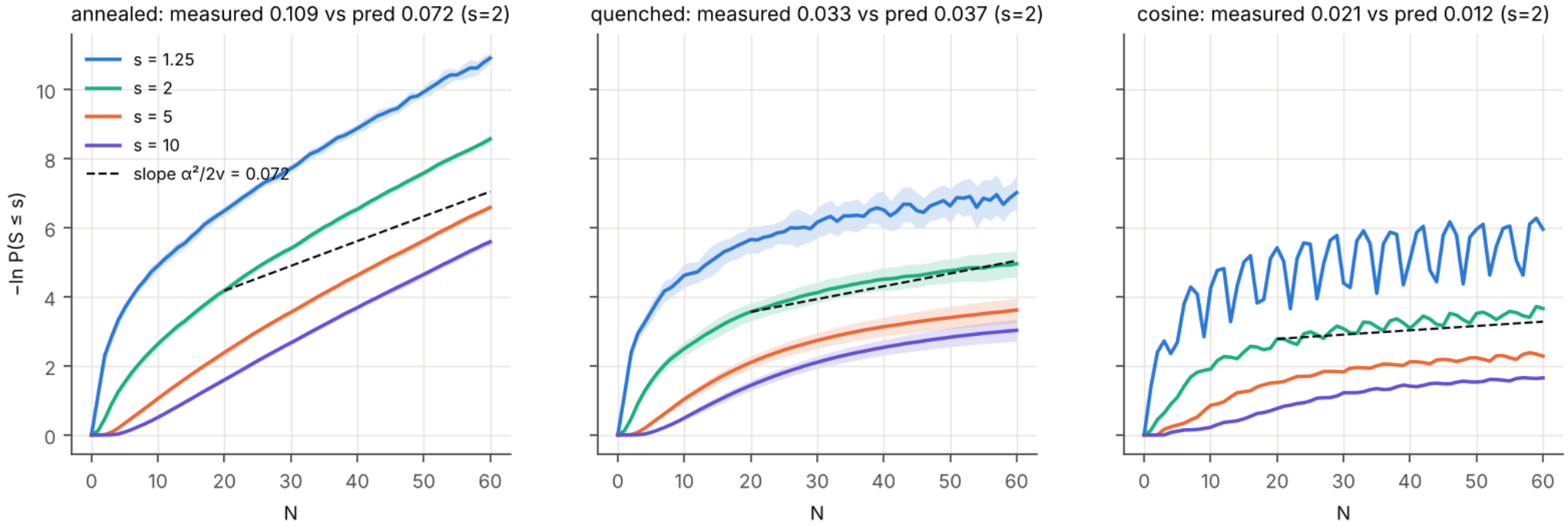
Distribution of S, quenched.



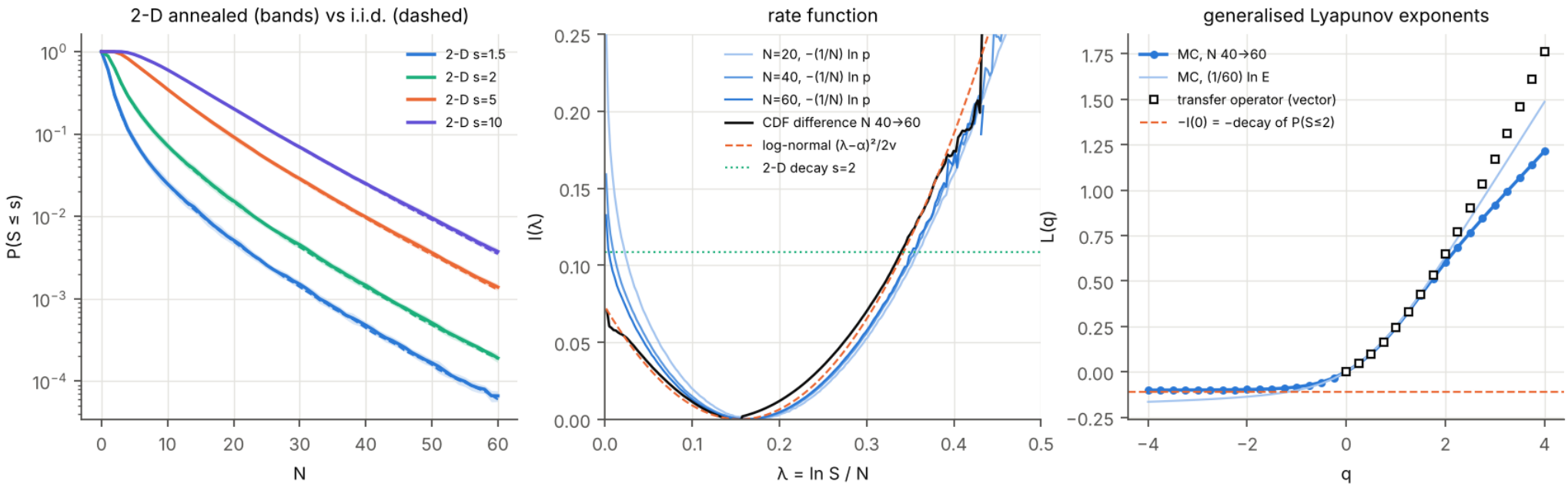
Distribution of S, cosine.



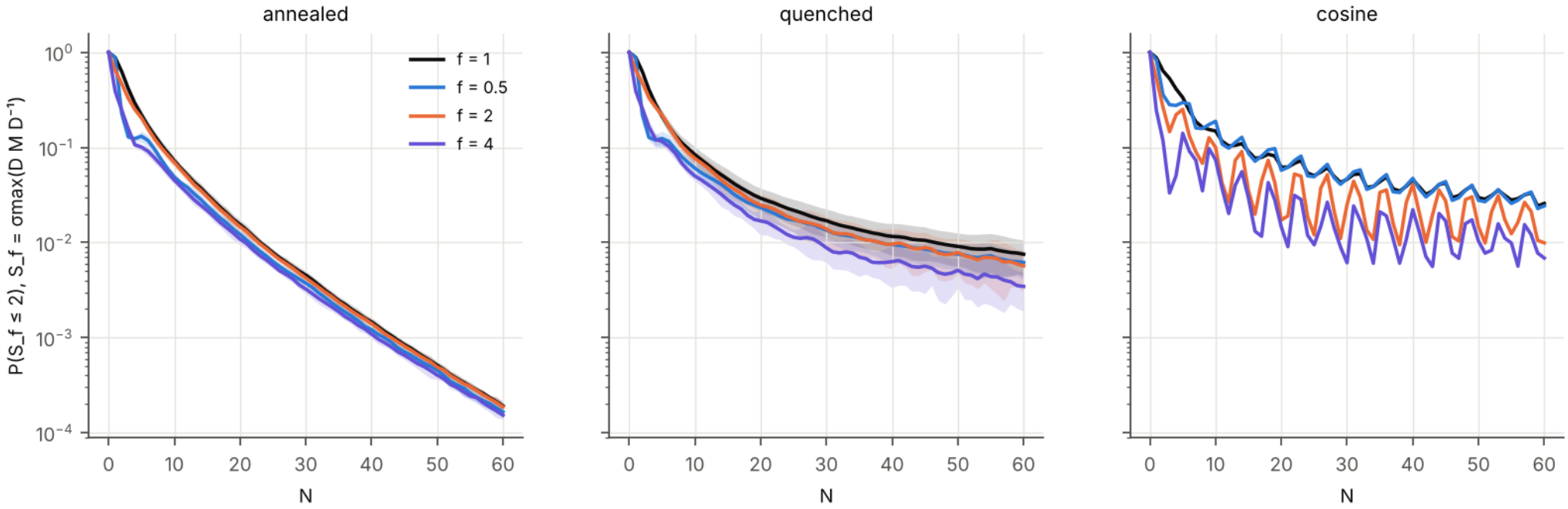
Collapse of (ln S − mean)/sd against a standard normal (seed 0).



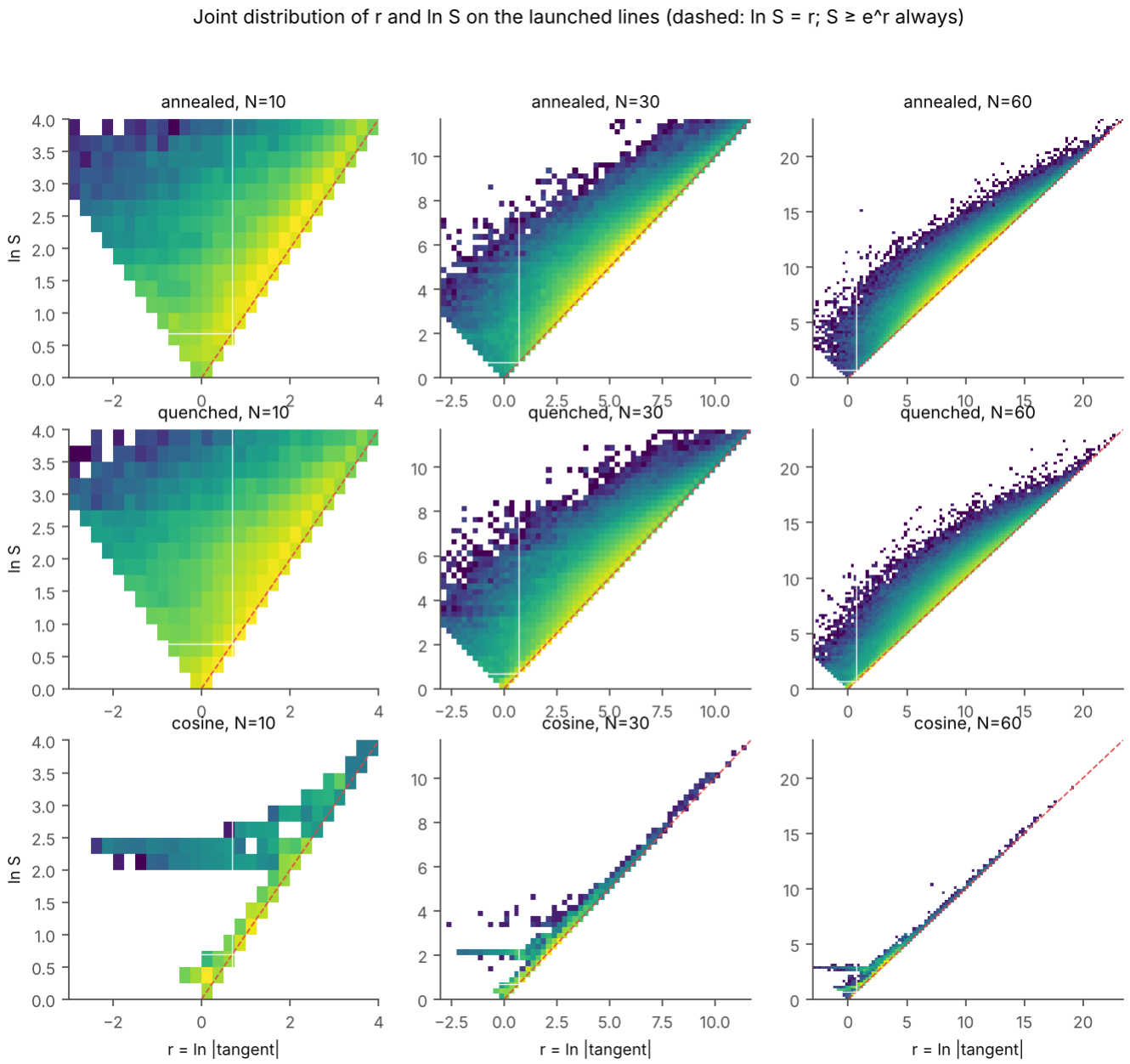
Kaplan log-normal tail test: −ln P(S≤s) vs N with the predicted slope.



i.i.d. random-matrix model: P(S≤s,N), rate function, L(q).



Metric rescaling y → f·y: threshold fractions move, decay rates do not.



Joint distribution of r vs ln S.

Method

Map. Yc = y + f(x), Xc = x + Tau·(Yc − row0), with f the piecewise-linear interpolant of the kick dp_x (rows = cols = 1920, K = 48, Tau = 0.5, row0 = 960, correlation length 96 px). M = [a b; c d] (rows x, y) is carried exactly by M ← J M, J = [1 + Tau·fp, Tau; fp, 1], det J = 1; S = σmax(M) = (√(t+2) + √(t−2))/2, t = a² + b² + c² + d².

Randomness. rng(s, 'twister') is emulated with MT19937 (seed 0 → 5489), and the stream is consumed in the script's order: per cycle the kick (959 uniforms) and then the dot jitter (2 × 25600). Quenched draws one kick before the loop; cosine uses no random numbers.

Lattices. The map is exactly periodic in y with period cols/Tau = 3840. One Lagrangian lattice seeded every 2 px over a full period ($1.84 \cdot 10^6$ parcels) gives (i) the MATLAB tracer lattice as its window-seeded subset (921,600 parcels; Parts 1B and 2) and (ii) an exact, uniform sample of the Eulerian window by selecting parcels whose y mod 3840 lies in [0, 1920) at step N (Part 4), in O(N) instead of the O(N²) backward sweep. Validated against MATLAB's 1-px Eulerian lattice: P(S ≤ s) and the ln S moments agree to 10^{-4}.

Lines (1A). Rows row0 and row0 ± rows/8, 8 parcels/px over all x, carried with M from I. Tangent (a, c); r = ln √(a² + c²). ρ(x, N) from the histogram (4 px bins, normalised so ρ = 1 at N = 0) and from Σ 1/|a| at bin centres along resolved segments (check on a 512/px line). Because the kick is piecewise linear, one step is affine inside each 1-px cell, so a is constant on pieces of the line and every caustic is a corner fold; the line is tracked exactly as pieces, cut where they cross integer x.

Crossing ages (2). Per tracer: cycle of last exit, cycle of last entry (0 = original cohort), number of returns and S at N−1. Risk and event counts are accumulated in [crossing cycle × age] arrays (and [returned before × depth × age]), from which every hazard, Kaplan–Meier curve and ratio follows.

Distribution of S (4). log10 S histogram in 0.05-decade bins on 0–13 plus overflow, P(S ≤ s), moments and quantiles of ln S and band shares, per N. The i.i.d. model draws fp from the first differences of 500 fresh kicks (960,000 values) and multiplies 10^7 independent products to N = 60.

Seeds. 10 seeds per mode, mean ± sd across seeds; figures show ±1 sd bands.

Caveats

• Seeds ≥ 1 assume MATLAB's rng(s, 'twister') uses MT19937 init_genrand(s) (as numpy RandomState(s); first draw for s = 1 is 0.4170). Seed 0 is MATLAB's documented stream. Kicks agree with MATLAB's to $\sim 10^{-7}$ relative: randpot1d_getFFT takes √|FFT(c)|, and in the Gaussian tail |FFT(c)| ~ 10^{-14} is round-off that differs between FFT libraries. Individual chaotic parcels can therefore differ at large N; statistics do not.

• Cosine mode is deterministic: all 10 seeds are the same run (sd = 0), not independent samples.

• Censoring: hazards need ≥ 200 at risk, so the whole-run Kaplan–Meier value is at the largest such age, not at k = 59. The naive fractions ignore censoring and are biased low for long ages.

• Part 2 follows material (Lagrangian), Part 4 measures the Eulerian window. The window is half a momentum period and is not invariant, so the two are different populations once parcels leave it.

• S depends on the coordinate metric (4E): thresholds such as S ≤ 2 and every probability built on them change with the scaling of y; growth rates (α) do not. Kaplan's test compares rates, but the prefactors behind the measured slopes are metric dependent at finite N.

• Caustic counts are a property of the piecewise-linear kick at 1-px resolution (corner folds); with a smooth kick the folds would be smooth and ρ would diverge as d^{−1/2}. Exact counts stop where the line exceeds $1.2 \cdot 10^7$ pieces (annealed N ≈ 31, quenched ≈ 33–35, cosine ≈ 41–44), so growth rates are fitted over N = 10 up to that point.

• "Type 1" is taken as the immediate bounce-back P(in the set at N | left at N−1), counted from consecutive masks; the check verifies the age bookkeeping reproduces it exactly.

• Finite N: 4C fits use N = 20..60; the Gaussian prediction is a central-limit statement while P(S ≤ s) probes the far lower tail, which large-deviation theory (4D) governs.